POLITECNICO DI TORINO

Mechanical Engineering

Course: Industrial Plants and Safety

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Industrial Plants and Safety

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1 INTRODUCTION

Alfa SpA is a company that works in the agricultural machinery sector and is planning to build a new factory in Northern Italy. The goal of this project is to design a complete plant that can produce and manage the logistics of two product families: **Flail Mowers** and **Rotary Tillers**. These machines are used in agriculture for soil preparation and cutting plant residues, and each family includes two sizes based on working width (L=1500 mm and L=1750 mm).

This report shows the decisions taken to organize the factory layout, production areas, handling systems, and warehouse zones. The factory must meet the required production volume for both product lines, following the given instructions and constraints. All steps of the design include technical reasoning and calculations, such as cycle times, takt time, and operator needs.

<u>The main points considered in this study are:</u>

- The factory must work efficiently to meet monthly production targets.
- All workstations must be balanced based on time and performance.
- Material flows should be optimized to reduce transport and waiting times.
- Warehouses must be sized to store components, semi-finished parts, and final products.
- A basic safety analysis must be done for manual handling of loads.

This report will include layout drawings, production flow diagrams, and detailed calculations to support the final proposal. The goal is to design a practical and functional

factory that meets the company's needs in a realistic and efficient way.

2 PRODUCT, PROCESS AND FLOWS

2.1 Required Production

To begin the plant design, we first calculated how many Flail Mowers and Rotary Tillers must be produced each month. Each product line includes two models: one with a working width of 1500 mm and one with 1750 mm. The required production depends on a formula given in the instructions:

$$P_i = 1.5 * \frac{C_m + N_m}{30} * Daily\ Base\ Production$$

Where P_i is the Total Required Output per Day

Where for Rotary Tiller:

$$C_m = 8\ and\ N_m = 6$$

And for Flail Mowers:

$$C_m = 13\ and\ N_m = 12$$

- C_m is the average value of the first letters of the last names of the group members (A=1, B=2, etc)
- N_m is the average value of the first letters of the first names of the group members. Once the values of C_m and N_m are inserted into the Excel sheet, the monthly production for each model is calculated automatically. This number is then used to size all departments in the factory (welding, bending, stamping, assembly, storage, etc)

  Daily base production is provided in the Excel file:

  - Rotary Tiller (L=1500) = 7352
  - Rotary Tiller (L=1750) = 3151
  - Flail Mower (L=1500) = 9389
  - Flail Mower (L=1750) = 6260

2.2 Bill of Materials (BOM)

Each product is made of several components, organized into levels from raw material to finished product. Most components are internally manufactured, while others are purchased from external suppliers.

The Bill of Materials (BOM) gives a full list of parts needed, grouped by model (L=1500 or L=1750). The usage rate shows how many of each component are needed to build the next level.

For example:

- To make one Guard L=1500, 0.17 of a metal sheet 1500x2000x3 is needed.
- To make one complete Flail Mower, several components like the welded frame, gearbox, shaft with hammers, guards, and hitch brackets are assembled.

The same logic applies to the Rotary Tillers. Some components are shared between both product lines and both model sizes.

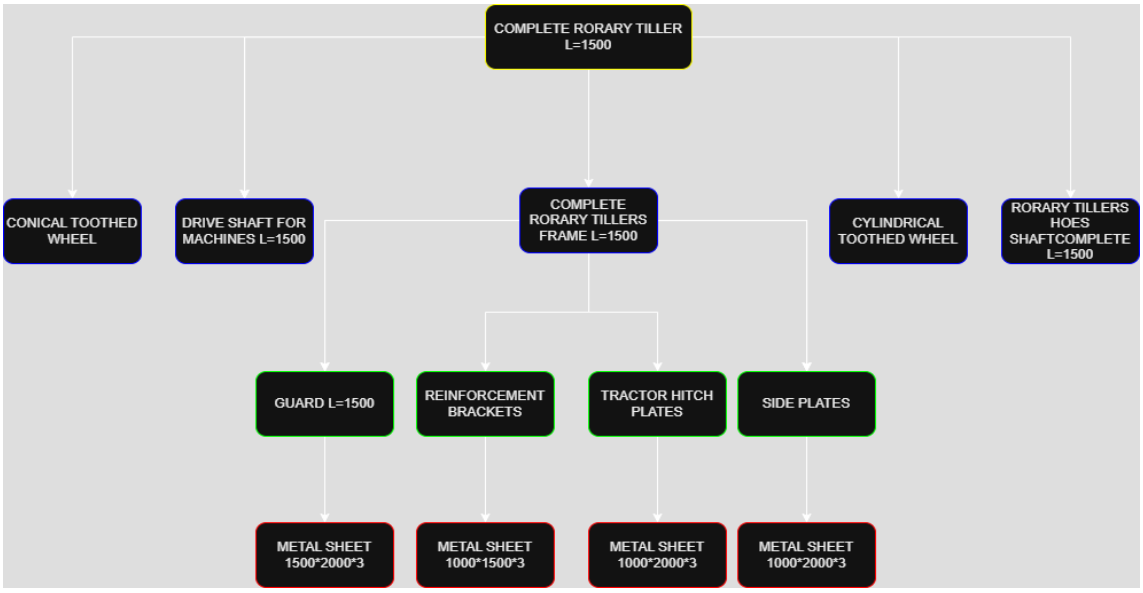


Figure 1:Rotary Tiller L=1500 BOM Tree

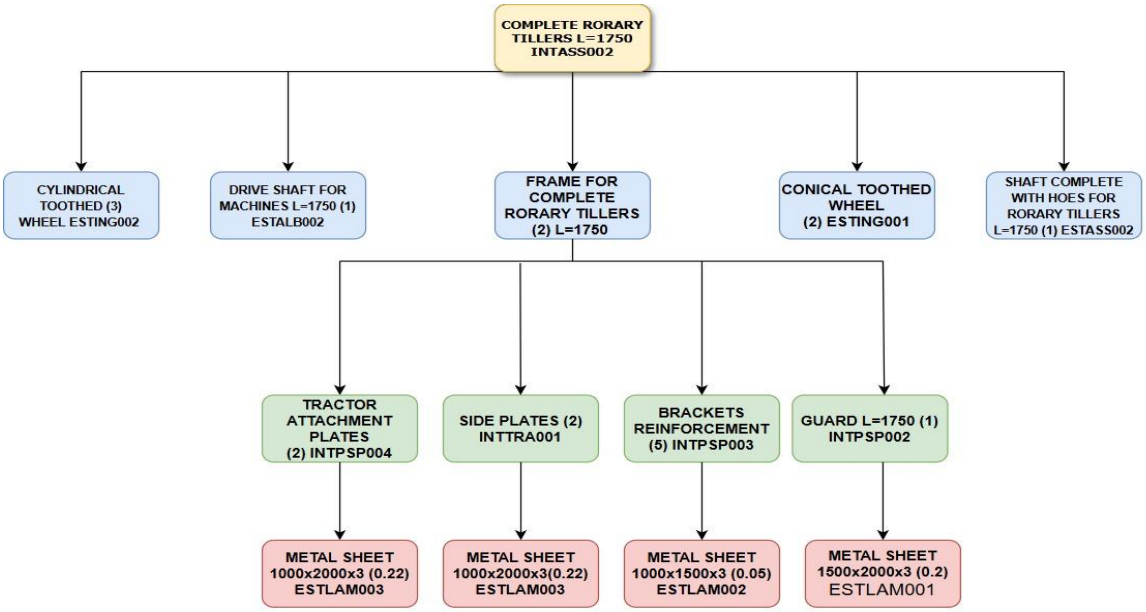


Figure 2:Rotary Tiller L=1750 BOM Tree

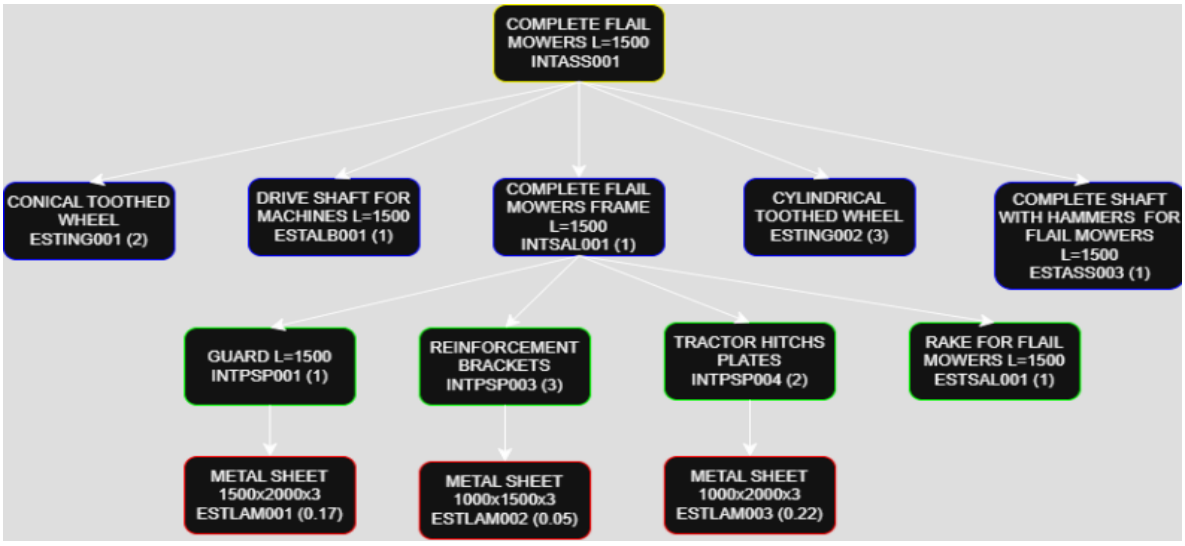


Figure 3:Flail Mower L=1500 BOM Tree

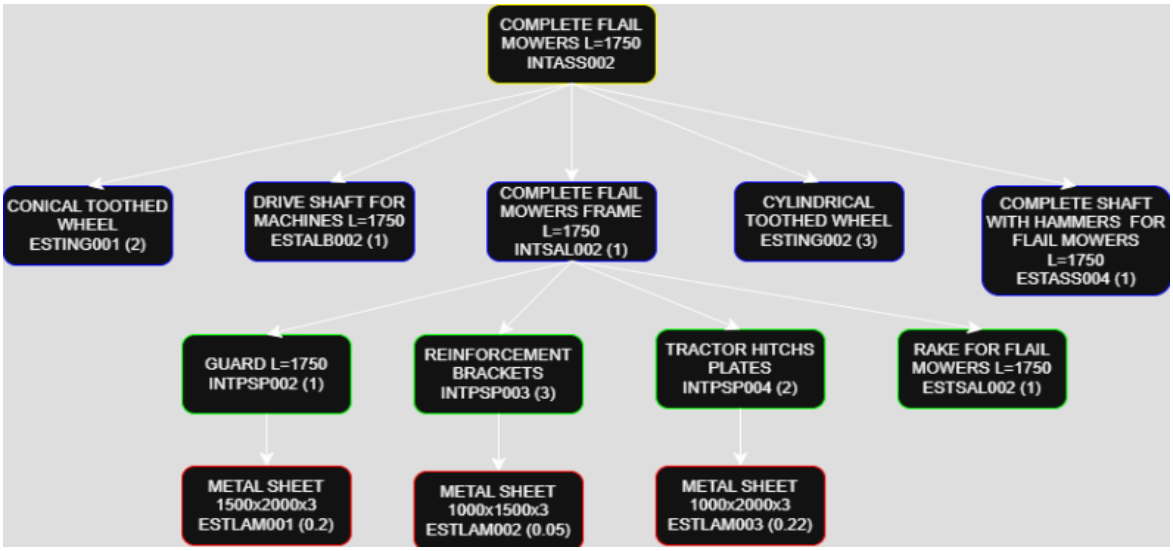


Figure 4:Flail Mower L=1750 BOM Tree

2.3 Material Flow

The production process for both **Flail Mowers** and **Rotary Tillers** includes multiple steps, starting from raw materials and ending with finished machines. The full path of components can be divided into different levels of transformation and assembly. Accounting for the; Availability, Performance and Quality rate of each process.

COD.	ROTARY TILLERS WORKCENTERS	AVAILABILITY (A)	PERFORMANCE (P)	QUALITY (Q)
ASS_FIN	ASSEMBLING LINE	97%	99%	100%
WELD	WELDING DEPARTMENT	98%	97%	97%
PSPG	BENDING DEPARTMENT	98%	99%	98%
TRNC	STAMPING DEPARTMENT	98%	99%	99%

Table 1:OEE Index Table for Rotary Tiller Production Line

COD.	FLAIL MOWERS WORKCENTERS	AVAILABILITY (A)	PERFORMANCE (P)	QUALITY (Q)
ASS_FIN	FINAL ASSEMBLING	97%	99%	100%
WELD	WELDING DEPARTMENT	98%	97%	97%
PSPG	BENDING DEPARTMENT	98%	99%	98%
TRNC	STAMPING DEPARTMENT	98%	99%	99%

Table 2:OEE Index Table for Flail Mower Production Line

The third step involves using the usage rate of the components to derive the required input to ensure that the desired output is met. To ensure this, we use the following equation:

$$Quantity\ of\ Products = Quantity\ of\ Products\ in\ Upper\ Level * Usage\ Rate$$

						CODE	TYPE	DESCRIPTION	USAGE RATE	Q.TY	Q.TY WITH SCRAPS
1						INTASS001	PART	**COMPLETE RORARY TILLERS L=1500**		7,352	
						INTASS001-10	OP	COMPLETE RORARY TILLERS ASSEMBLY L=1500			7,352
	2					INTSAL001	PART	**COMPLETE RORARY TILLERS FRAME L=1500**	1	7,352	
						INTSAL001-10	OP	WELDING COMPLETE FRAME WITH TRACTOR HITCH L=1500			7,579
		3				INTPSP001	PART	GUARD L=1500	1	7,579	
						INTPSP001-20	OP	GUARD BENDING L=1500			7,734
						INTPSP001-10	OP	GUARD STAMPING L=1500			7,812
			4			ESTLAM001	PART	METAL SHEET 1500x2000x3	0.17	1,562	
		3				INTPSP003	PART	REINFORCEMENT BRACKETS	3	22,738	
						INTPSP003-20	OP	REINFORCEMENT BRACKETS BENDING			23,203
						INTPSP003-10	OP	REINFORCEMENT BRACKETS STAMPING			23,437
			4			ESTLAM002	PART	METAL SHEET 1000x1500x3	0.05	1,172	
		3				INTPSP004	PART	TRACTOR HITCH PLATES	2	15,159	
						INTPSP004-20	OP	TRACTOR HITCH PLATES BENDING			15,468
						INTPSP004-10	OP	TRACTOR HITCH PLATES STAMPING			15,625
			4			ESTLAM003	PART	METAL SHEET 1000x2000x3	0.22	3,906	
		3				INTTRA001	PART	SIDE PLATES	2	15,159	
						INTTRA001-10	OP	SIDE PLATES STAMPING			15,312
			4			ESTLAM003	PART	METAL SHEET 1000x2000x3	0.22	3,828	
	2					ESTALB001	PART	**DRIVE SHAFT FOR MACHINES L=1500**	1	7,579	
	2					ESTING001	PART	**CONICAL TOOTHED WHEEL**	2	15,159	
	2					ESTING002	PART	**CYLINDRICAL TOOTHED WHEEL**	3	22,738	
	2					ESTASS001	PART	**RORARY TILLERS HOES SHAFT COMPLETE L=1500**	1	7,579	
1						INTASS002	PART	**COMPLETE RORARY TILLERS L=1750**		3,151	
						INTASS002-10	OP	COMPLETE RORARY TILLERS ASSEMBLY L=1750			3,151
	2					INTSAL002	PART	**FRAME FOR COMPLETE RORARY TILLERS L=1750**	1	3,151	
						INTSAL002-10	OP	WELDING COMPLETE FRAME WITH TRACTOR HITCHS L=1750			3,248
		3				INTPSP002	PART	GUARD L=1750	1	3,248	
						INTPSP002-20	OP	GUARD BENDING L=1750			3,315
						INTPSP002-10	OP	GUARD STAMPING L=1750			3,348
			4			ESTLAM001	PART	METAL SHEET 1500x2000x3	0.20	670	
		3				INTPSP003	PART	REINFORCEMENT BRACKETS	5	16,242	
						INTPSP003-20	OP	REINFORCEMENT BRACKETS BENDING			16,573
						INTPSP003-10	OP	REINFORCEMENT BRACKETS STAMPING			16,741
			4			ESTLAM002	PART	METAL SHEET 1000x1500x3	0.05	837	
		3				INTPSP004	PART	TRACTOR ATTACHMENT PLATES	2	6,497	
						INTPSP004-20	OP	TRACTOR HITCH PLATES BENDING			6,629
						INTPSP004-10	OP	TRACTOR HITCH PLATES STAMPING			6,696
			4			ESTLAM003	PART	METAL SHEET 1000x2000x3	0.22	1,674	
		3				INTTRA001	PART	SIDE PLATES	2	6,497	
						INTTRA001-10	OP	SIDE PLATES STAMPING			6,562
			4			ESTLAM003	PART	METAL SHEET 1000x2000x3	0.22	1,641	
	2					ESTALB002	PART	**DRIVE SHAFT FOR MACHINES L=1750**	1	3,151	
	2					ESTING001	PART	**CONICAL TOOTHED WHEEL**	2	6,302	
	2					ESTING002	PART	**CYLINDRICAL TOOTHED WHEEL**	3	9,453	
	2					ESTASS002	PART	**SHAFT COMPLETE WITH HOES FOR RORARY TILLERS L=1750**	1	3,151	

Table 3:Flow of Material of Rotary Tiller

Moreover, taking into account the quality of each in the Assembly and Welding departments, the Net Quantity, which is the quantity required to achieve the desired demand output, will increase respecting the relative proportion of the rejection rate and we would get the Gross quantity which is the quantity that needs to be taken into account in the whole production line, anticipating the failure of some components.

					CODE	TYPE	DESCRIPTION	USAGE RATE	Q.TY (pcs/month)	Q.TY WITH SCRAP (pcs/month)
BILL OF MATERIAL LEVELS					CODE	TYPE	DESCRIPTION	USAGE RATE	Q.TY	Q.TY WITH SCRAPS
1					INTASS001	PART	**COMPLETE FLAIL MOWERS L=1500**		9,389	
					INTASS001-10	OP	COMPLETE FLAIL MOWERS ASSEMBLY L=1500			**9,389**
	2				INTSAL001	PART	**COMPLETE FLAIL MOWERS FRAME L=1500**	1	9,389	
					INTSAL001-10	OP	WELDING FRAME WITH TRACTOR HITCHS L=1500			**9,679**
		3			INTPSP001	PART	GUARD L=1500	1	9,679	
					INTPSP001-20	OP	GUARD BENDING L=1500			**9,877**
					INTPSP001-10	OP	GUARD STAMPING L=1500			**9,777**
			4		ESTLAM001	PART	METAL SHEET 1500x2000x3	0.17	1,955	
		3			INTPSP003	PART	REINFORCEMENT BRACKETS	3	29,331	
					INTPSP003-20	OP	REINFORCEMENT BRACKETS BENDING			**29,930**
					INTPSP003-10	OP	REINFORCEMENT BRACKETS STAMPING			**30,232**
			4		ESTLAM002	PART	METAL SHEET 1000x1500x3	0.05	1,512	
		3			INTPSP004	PART	TRACTOR HITCHS PLATES	2	18,778	
					INTPSP004-20	OP	TRACTOR HITCH BRACKETS BENDING			**19,161**
					INTPSP004-10	OP	TRACTOR HITCHS BRACKETS STAMPING			**19,355**
			4		ESTLAM003	PART	METAL SHEET 1000x2000x3	0.22	4,839	
		3			ESTSAL001	PART	RAKE FOR FLAIL MOWERS L=1500	1	9,389	
	2				ESTALB001	PART	**DRIVE SHAFT FOR MACHINES L=1500**	1	9,389	
	2				ESTING001	PART	**CONICAL TOOTHED WHEEL**	2	18,778	
	2				ESTING002	PART	**CYLINDRICAL TOOTHED WHEEL**	3	28,167	
	2				ESTASS003	PART	**COMPLETE SHAFT WITH HAMMERS FOR FLAIL MOWERS L=1500**	1	9,389	
1					INTASS002	PART	**COMPLETE FLAIL MOWERS L=1750**		6,260	
					INTASS002-10	OP	COMPLETE FLAIL MOWERS ASSEMBLY L=1750			**6,260**
	2				INTSAL002	PART	**COMPLETE FLAIL MOWERS FRAME L=1750**	1	6,260	
					INTSAL002-10	OP	WELDING FRAME. WITH TRACTOR HITCHS L=1750			**6,454**
		3			INTPSP002	PART	GUARD L=1750	1	6,260	
					INTPSP002-20	OP	GUARD BENDING L=1750			6,388
					INTPSP002-10	OP	GUARD STAMPING L=1750			**6,452**
			4		ESTLAM001	PART	METAL SHEET 1500x2000x3	0.20	1,290	
		3			INTPSP003	PART	REINFORCEMENT BRACKETS	5	31,300	
					INTPSP003-20	OP	REINFORCEMENT BRACKETS BENDING			31,939
					INTPSP003-10	OP	REINFORCEMENT BRACKETS STAMPING			32,261
			4		ESTLAM002	PART	METAL SHEET 1000x1500x3	0.05	1,613	
		3			INTPSP004	PART	TRACTOR HITCHS PLATES	2	12,520	
					INTPSP004-20	OP	TRACTOR HITCH BRACKETS BENDING			12,776
					INTPSP004-10	OP	TRACTOR HITCHS BRACKETS STAMPING			12,905
			4		ESTLAM003	PART	METAL SHEET 1000x2000x3	0.22	3,226	
		3			ESTSAL002	PART	RAKE FOR FLAIL MOWERS L=1750	1	6,260	
	2				ESTALB002	PART	**DRIVE SHAFT FOR MACHINE L=1750**	1	6,260	
	2				ESTING001	PART	**CONICAL TOOTHED WHEEL**	2	12,520	
	2				ESTING002	PART	**CYLINDRICAL TOOTHED WHEEL**	3	18,780	
	2				ESTASS004	PART	**COMPLETE SHAFT WITH HAMMERS FOR FLAIL MOWERS L=1750**	1	6,260	

Table 4:Flow of Material of Flail Mower

In what concerns the Flail Mower, the same procedure is followed considering the different values of quality and rejections rates.

2.4 Production Process and Overall Flows

A process chart and flow diagram are prepared to show all material movements and manufacturing steps. Each item in the BOM is traced through its full processing steps, including the materials moved in Load Units (LU) of a specified component through processes and warehouses.

FROM (row) / TO (column)	RM WAREHOUSE	FP WAREHOUSE	FINAL ASSEMBLING	WELDING BUFFER	WELDING	BENDING BUFFER	BENDING	STAMPPING BUFFER	STAMPING
RM WAREHOUSE									
FP WAREHOUSE			2,227						
FINAL ASSEMBLING	1,380			847					
WELDING BUFFER					847				
WELDING						738		109	
BENDING BUFFER							738		
BENDING								760	
STAMPING BUFFER									869
STAMPING	393								

INPUT PER MONTH [LU/MONTH] IN RM WAREHOUSE	METAL SHEET	GEARS	DRIVE SHAFT	COMPLETE SHAFT WITH HOES FOR FLAIL MOWERS
	393	269	216	895

OUTPUT PER MONTH [LU/MONTH] FROM FP WAREHOUSE	ROTARY TILLER L=1500	ROTARY TILLER L=1750	Departures from the PF warehouse occur daily. The daily quantity is shipped.
	7,353	3,151	

Table 5:Flow of Material of Rotary Tiller Production Line

FROM (row) / TO (column)	RM WAREHOUSE	FP WAREHOUSE	FINAL ASSEMBLING	WELDING BUFFER	WELDING	BENDING BUFFER	BENDING	STAMPING BUFFER	STAMPING
RM WAREHOUSE									
FP WAREHOUSE			4,416						
FINAL ASSEMBLING	2,011			2,405					
WELDING BUFFER					2,405				
WELDING	1,305					1,100			
BENDING BUFFER							1,100		
BENDING								1,126	
STAMPING BUFFER									1,126
STAMPING	492								

INPUT PER MONTH [LU/MONTH] IN RM WAREHOUSE	METAL SHEET	GEARS	DRIVE SHAFT	COMPLETE SHAFT WITH HAMMER	RAKE FOR FLAIL MOWERS
	492	392	314	1,305	1,305

OUTPUT PER MONTH [LU/MONTH] FROM FP WAREHOUSE	FLAIL MOWERS L=1500	FLAIL MOWERS L=1750	Departures from the PF warehouse occur daily. The daily quantity is shipped.
	9,389	6,260	

Table 6:Flow of Material of Flail Mower Production Line

The chart also considers quality rates and rejection rates from the OEE table. This ensures we calculate not only the net quantity, but also the gross quantity to account for production losses.

Below is a simplified sketch of the process' and the Load Units representation. In the following slides, a more in-depth sketch can be found demonstrating the details.

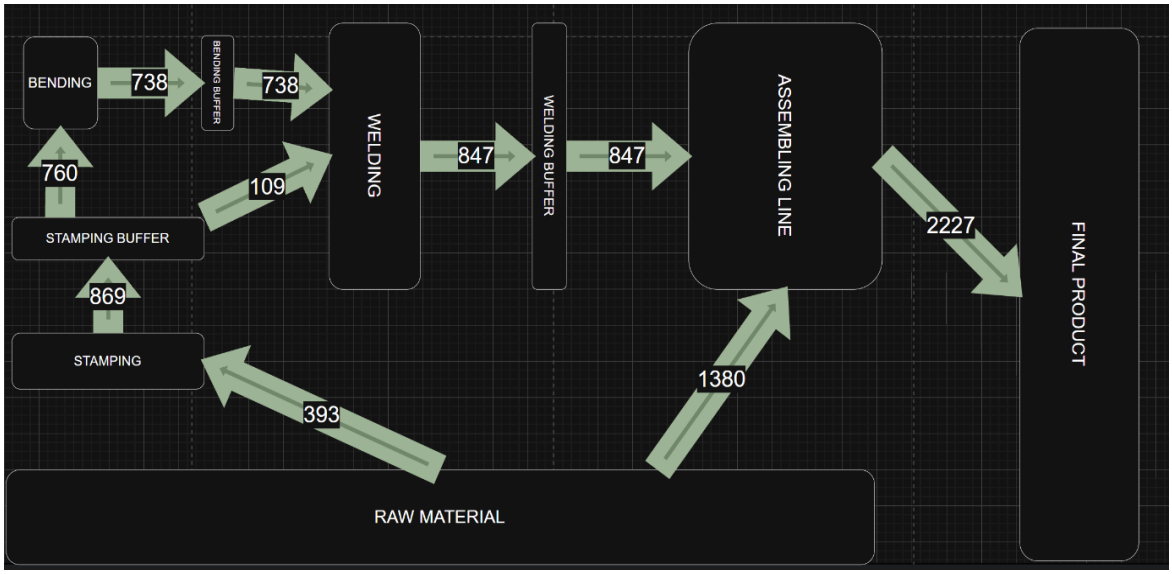


Figure 1:Diagram of moved material in Rotary Tiller Production Line

For Rotary Tiller, the material flow starts from the Raw Material Warehouse and moves to stamping. Afterwards, most parts are sent to Bending, however some of them go straight to welding as no bending is necessary. Then the parts proceed to a short welding buffer before they proceed to assembling line where all remaining components (like shafts, covers and brackets) are added with the parts coming straight from the Raw Material Warehouse. This is how the Rotary Tiller is made.

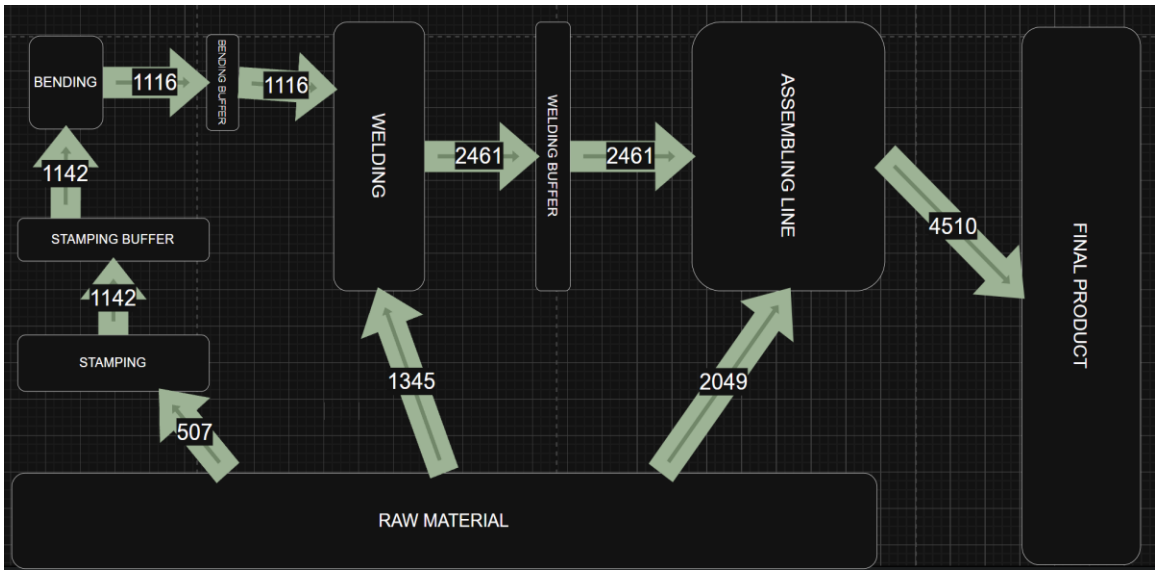


Figure 2:Diagram of moving material in Flail Mower Production Line

For Flail Mower, process is similar to Rotary Tiller; except rather than sending some parts from stamping buffer to welding, we send them straight to welding without even being stamped. The rest of the process is the same other than the load units.

3 Layout Type

3.1 Choice for Layout type

The layout selection process began by analyzing the cycle times of the different departments (Assembling, welding, bending-stamping). For both the Flail MOWER and Rotary Tiller, welding takes **8 minutes** per unit while assembly takes **2 minutes**, bending and stamping take **1 minute**. Since welding has a longer cycle time for both products, it creates a bottleneck. A line layout would reduce plant availability (A) and overall equipment effectiveness (OEE) due to uneven workflow, whereas a process layout allows flexible scaling of welding stations to minimize downtime and balance production efficiency.

3.2 Assigning Workstations in Different Departments

OEE is a vital tool for enhancing the operational efficiency of an industrial plant, enabling better utilization of assets, improving production processes, and ultimately contributing to increased profitability.

In our plant, one of our main goals is **increasing the OEE** as much as possible working with resources in hand:

-We began by analyzing Total Production Time and output quantities for welding and assembly lines across both products.

-Then derived Takt Time as the ratio between production time and total production in pieces.

-Additionally, after dividing the cycle time, which is the theoretical time permitted to manufacture a certain output, by the Takt Time we would have a theoretical number of

Lines (for assembly).

-In parallel, a **successor diagram** is put together for each product in order to better understand the sequence of tasks and assign workstations dedicated to a certain group of tasks.

Relatively our task flow is simple and linear and thus the process of assigning groups of tasks together would be trivial. In this process, **RPW** for each task is calculated and tasks are assigned in descending order, followed by constraining the maximum time of a certain group of tasks by the cycle time and finally deriving the actual number of workstations needed.

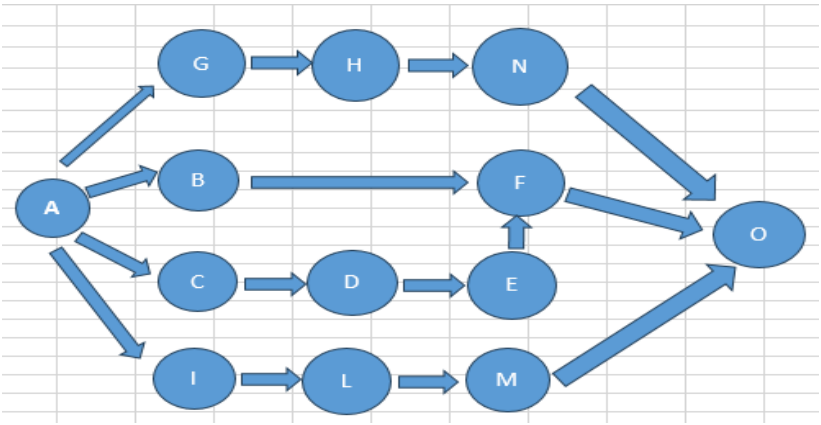


Figure 7:Flail Mower successor diagram

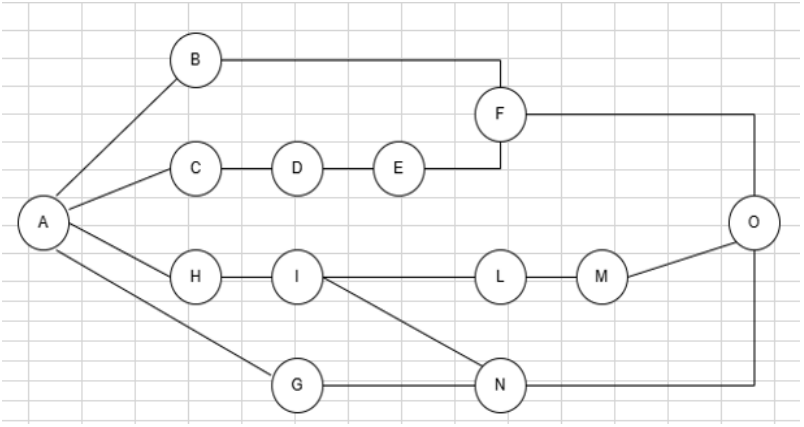


Figure 8:Rotary Tiller successor diagram

Our calculations resulted in a total of **4 workstations** for each the Flail Mower and the Rotary Tiller.

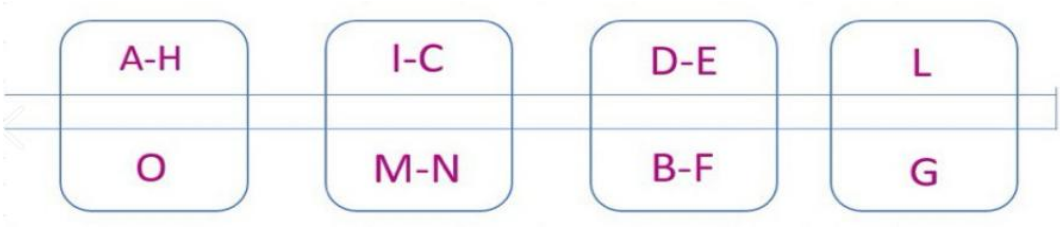


Figure 9:Preliminary design of the final shape of the Rotary Tiller assembly.

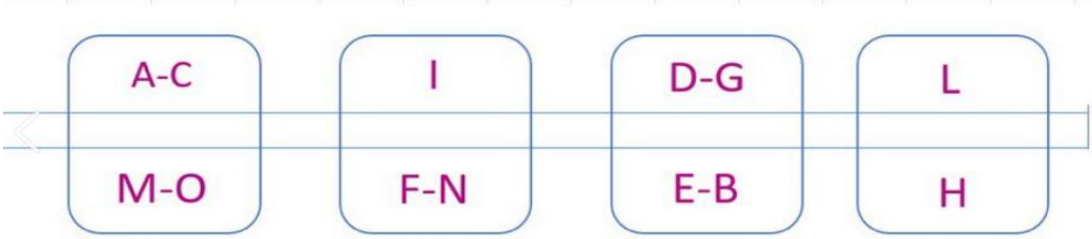


Figure 10:Preliminary design of the final shape of the Flail Mower assembly.

3.3 – Workstation Layout

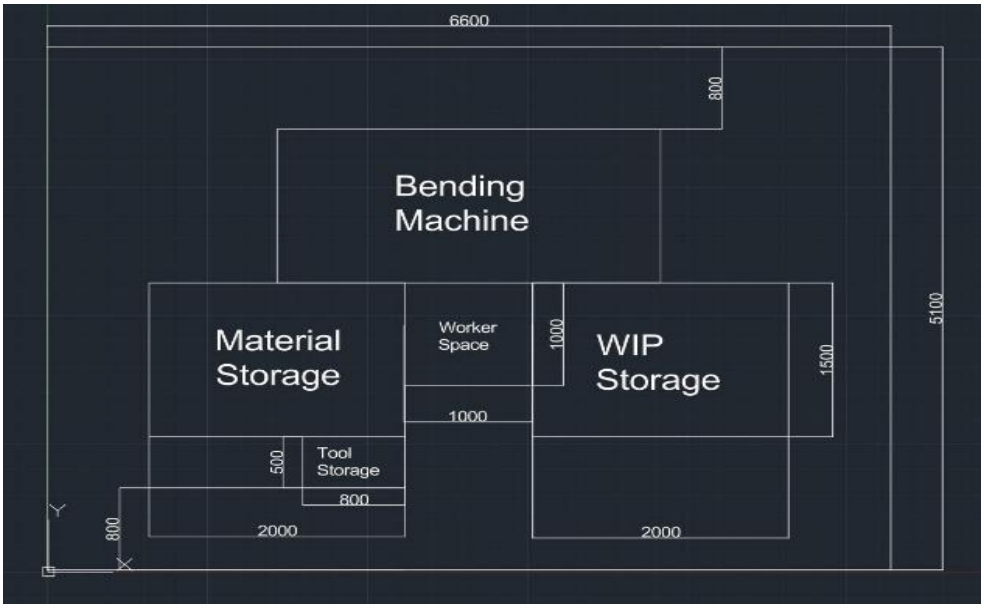


Figure 11:bending workstation layout

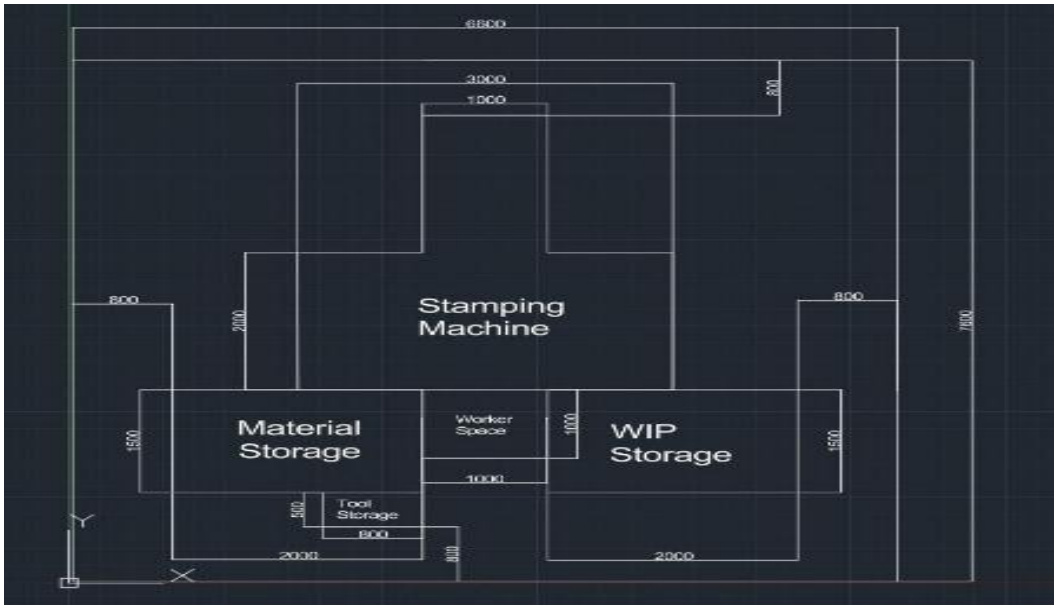


Figure 12:stamping workstation layout

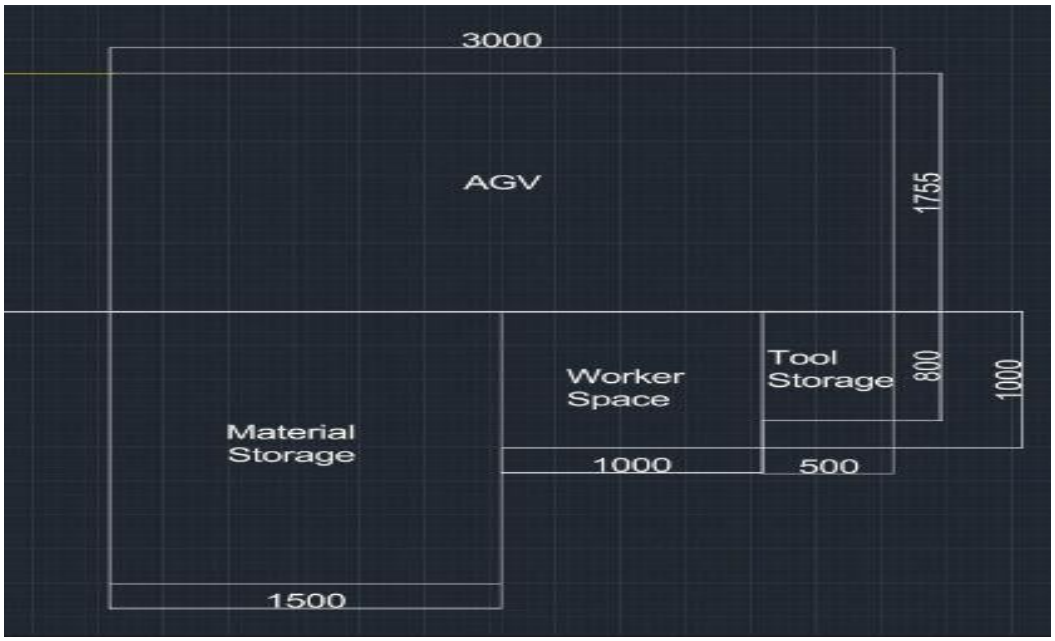


Figure 13:Assembly workstation layout

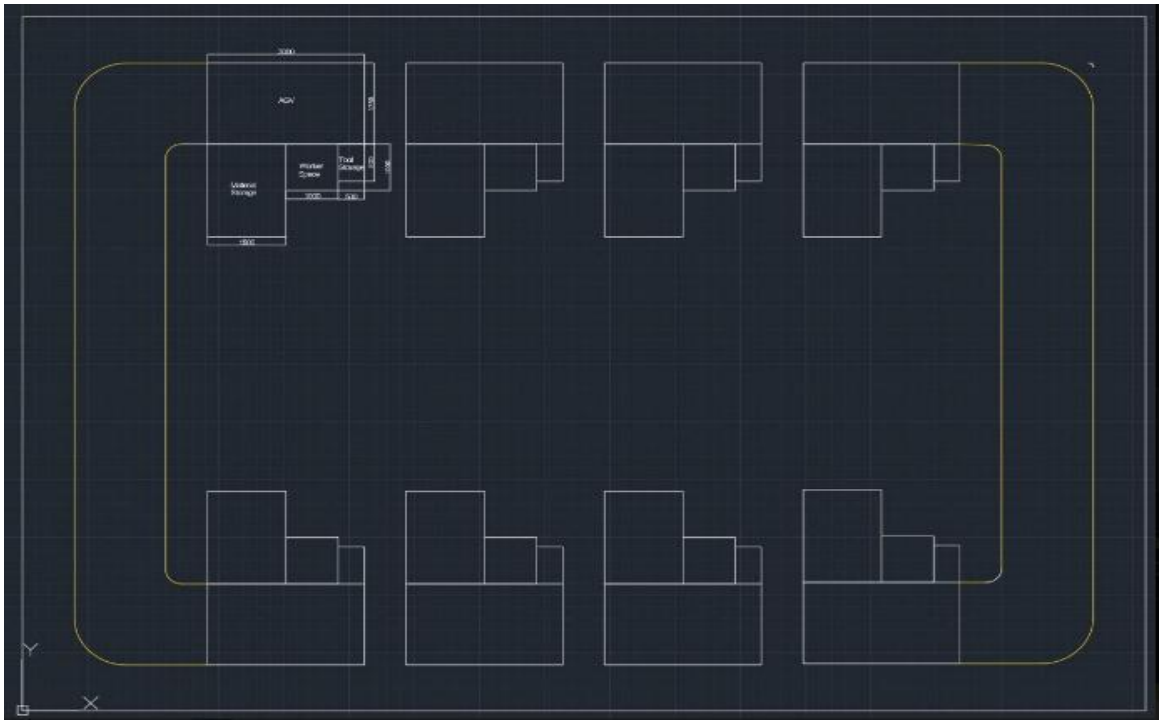


Figure 14: Assembly Line

3.4 Assembly Task for FLAIL MOWER

Workstation 1: is critical to establishing a stable and functional base for further assembly.

1. Task 1: Take frame assembly from load unit and place on equipment. This forms the foundational structure on which all other components will be mounted.

2. Task 2: Take gear box and fasten on chassis. This provides a stable base for the motor, ensuring it is securely fixed to the frame.

3. Task 3: Pick up register hammer cover and assemble on frame This step ensures the motor is firmly attached, providing the necessary power for the Flail Mower operation.

4. Task 4: Take complete flail mower and lay down in load unit, this task prepares it for the installation.

Workstation 2: focuses on ensuring the motor is safeguarded.

1. Task 1: Take shaft with hammers and assemble on frame.

2. Task 2: Take drive shaft and assemble

3. Task 3: Remove side gear cover and assemble on frame

Workstation 3:

1. Task 1: Take scrapers and fasten on gear box.

2. Task 2: Take frame side closure and assemble.

3. Task 3: Take bevel gear pair and assemble within box

4. Task 4: Take drive shaft supports and secure on chassis

Workstation 4:

1. Task 1: Remove hammers cover and assemble on frame.

2. Task 2: Take out gears and mount on frame side closure.

4 Material Handling

4.1 Type of Storage

For our plant, we have decided to use **3 different types of storage**:

1- **Live storage**: Utilizes vertical space to maximize storage capacity within the same footprint taking advantage of our available height. Live storage acts as a buffer to absorb fluctuations in supply and demand, ensuring that production can continue even if there are temporary disruptions in the supply chain which increases the efficiency of our production and helps meeting the demand without interruption.

2- **Drive-THRU:** Drive-thru racking offers high-density storage by eliminating multiple aisles allowing forklifts to drive directly into the rack structure to store and retrieve pallets which is cost-effective.

3- **SELECTIVE PALLET RACKING**: Provides direct, individual access to every pallet, making it the most flexible system for warehouses. Unlike drive-thru or live storage, no pallet must be moved to reach another, enabling fast picking and restocking. The adjustable beam heights

accommodate varying load sizes, and it works seamlessly with forklifts. Its speed and adaptability make it ideal for fast-moving inventories, just-in-time operations, and facilities with frequent order changes.

For defining the warehouses, we grouped the material depending on the storage system inside each warehouse/buffer. Using this method, the warehouses can be divided into **5** parts:

1. **RAW MATERIALS WAREHOUSE:** Uses SELECTIVE PALLET RACKING
Has a total storage area equal to 398.512 m^2 in the case of rotary tiller, and 1230.97 m^2 in the case of Flail mower.

2. **STAMPING BUFFER WAREHOUSE**: Uses Drive-THRU
Has a storage area of 65.18 m^2

3. **BENDING BUFFER WAREHOUSE**: Uses Drive-THRU
Has a storage area of 65.18 m^2

4. **WELDING BUFFER WAREHOUSE**: Uses Drive-THRU
Has a storage area of 134.99 m^2

5. **FINAL PRODUCT WAREHOUSE**: Uses Live storage
Has a storage area of 1267.98 m^2.
The finished product warehouse is dedicated to storing completed products ready for distribution or shipment.

4.2 EPEI

Every Product Every Interval (EPEI) is a concept used in lean manufacturing and production scheduling to determine how often each product is produced within a given time frame. The goal of EPEI is to reduce inventory and lead time by producing smaller batches of each product more frequently.

This approach helps in maintaining a smooth production flow and responding more quickly to changes in demand. We should optimize our batch sizes using this method .

$$\frac{Total\ step\ time\ due\ to\ the\ product\ family}{Productive\ time - Machine\ time}$$

In our project for both Flail Mower and Rotary Tiller , we calculate the batches for the different operations (BENDING-STAMPING , WELDING and FINAL ASSEMBLY) following the same procedure , we have 2 designs (L=1500 and L=1750) so we should calculate the batches for each .

We follow these steps to get the EPEI :

1. We calculate the production per day per machine is:

$$\frac{Demand\ total\ quantity}{number\ of\ machines} \times lines \times 22$$

2. We calculate the run time that is:

$$Production\ per\ day\ per\ machine \times Cycle\ time$$

3. We get EPEI that is:

$$\frac{total\ setup\ time}{potential\ available\ time} \times Performance\ (\%) - Run\ time$$

4. EPEI batch for a single machine is:

$$Production\ per\ day\ per\ machine \times EPEI\ per\ day$$

5. Department EPEI batch is:

$$EPEI\ batch\ for\ single\ machine \times total\ number\ of\ machines$$

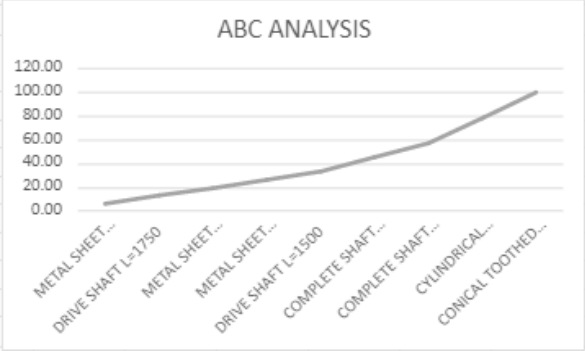
6. If we want to get the semi-finished product batch is:

Department EPEI batch × quality rate

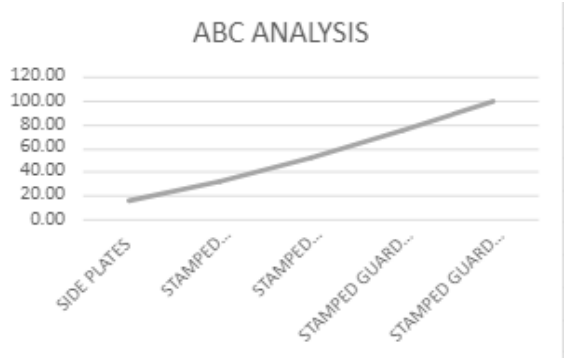
4.3 Logistic Performance Indices

ABC ANALYSIS for Rotary Tiller:

RAW MATERIAL WAREHOUSE ABC ANALYSIS			
Components	Turnover Index	Cummilative Turnover Index	Cummilative Turnover Index (%)
METAL SHEET 1500x2000x3	2.75	2.75	6.69
DRIVE SHAFT L=1750	2.75	5.50	13.37
METAL SHEET 1000x1500x3	2.75	8.25	20.06
METAL SHEET 1000x2000x3	2.75	11.00	26.75
DRIVE SHAFT L=1500	2.75	13.75	33.43
COMPLETE SHAFT WITH HOES FOR ROTARY TILLER L=1500	4.89	18.64	45.32
COMPLETE SHAFT WITH HOES FOR ROTARY TILLER L=1750	4.89	23.53	57.21
CYLINDRICAL TOOTHED WHEEL	8.80	32.33	78.60
CONICAL TOOTHED WHEEL	8.80	41.13	100

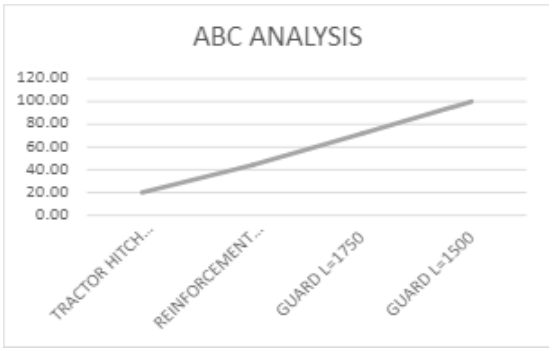


STAMPING BUFFER WAREHOUSE ABC ANALYSIS			
Components	Turnover Index	Cummilative Turnover Index	Cummilative Turnover Index (%)
SIDE PLATES	15.62	15.62	16.26
STAMPED TRACTOR HITCH PLATES	16.10	31.73	33.03
STAMPED REINFORCEMENT BRACKETS	19.33	51.05	53.14
STAMPED GUARD L=1750	22.47	73.53	76.53
STAMPED GUARD L=1500	22.55	96.07	100



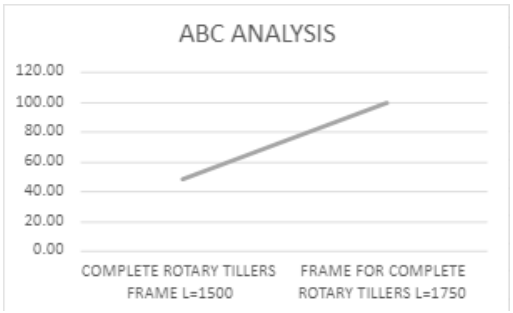
BENDING BUFFER WAREHOUSE ABC ANALYSIS

Components	Turnover Index	Cummilative Turnover Index	Cummilative Turnover Index (%)
TRACTOR HITCH PLATES	15.78	15.78	20.02
REINFORCEMENT BRACKETS	18.94	34.72	44.04
GUARD L=1750	22.02	56.75	71.97
GUARD L=1500	22.10	78.85	100



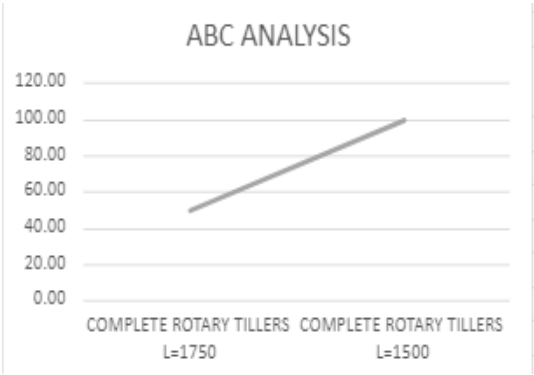
WELDING BUFFER WAREHOUSE ABC ANALYSIS

Components	Turnover Index	Cummilative Turnover Index	Cummilative Turnover Index (%)
COMPLETE ROTARY TILLERS FRAME L=1500	20.94	20.94	49.16
FRAME FOR COMPLETE ROTARY TILLERS L=1750	21.66	42.59	100



FINAL PRODUCT WAREHOUSE ABC ANALYSIS

Components	Turnover Index	Cummilative Turnover Index	Cummilative Turnover Index (%)
COMPLETE ROTARY TILLERS L=1750	20.94	20.94	49.92
COMPLETE ROTARY TILLERS L=1500	21.01	41.94	100



4.5 Picking Area

We have to design two picking areas, one for receiving the raw material and one for shipping the finished product. First, we need to schedule a weekly timetable for material arrival. Our idea is to uniformly distribute the LU received during the days of the week in order not to oversize unnecessarily the docks and receiving area. We distributed the received products on four weeks.

Week 1:

Monday	Tuesday	Wednesday	Thursday	Friday
RAKE FOR FLAIL MOWERS	**METAL SHEETS**	**METAL SHEETS**	**DRIVE SHAFTS**	**COMPLETE SHAFTS WITH HAMMERS**
	GEARS		**COMPLETE SHAFTS WITH**	**GEARS**

			HOES	
	GEARS		DRIVE SHAFTS	GEARS
LU 795	456	339	474	486

Week 2:

Monday	Tuesday	Wednesday	Thursday	Friday
		GEARS	COMPLETE SHAFTS WITH HOES	COMPLETE SHAFTS WITH HAMMERS
		GEARS		
LU		123	285	363

Week 3:

Monday	Tuesday	Wednesday	Thursday	Friday
RAKE FOR FLAIL MOWERS	METAL SHEETS	METAL SHEETS	DRIVE SHAFTS	COMPLETE SHAFTS WITH HAMMERS
GEARS			COMPLETE SHAFTS WITH HOES	
GEARS			DRIVE SHAFTS	
			GEARS	
			GEARS	
LU 918	333	339	725	363

Week 4:

Monday	Tuesday	Wednesday	Thursday	Friday
	GEARS		**COMPLET E SHAFTS WITH HOES**	**COMPLETE SHAFTS WITH HAMMERS**
	GEARS			**GEARS**
				GEARS
LU	123		285	486

In BLUE: Materials for Flail Mower.

In PINK: Materials for Rotary Tiller.

In RED: Highest number of load units received per day.

In order to be conservative, we oversize our docks and receiving area by taking into account the day with the height number of Load Units (Monday – 918 LU) and the greatest LU size (1x2x1). The truck chosen for transportation is a FEU (63.4mˆ3) truck.

The calculation includes:

• Quantity of LU/hour: In total, we have 1730 LU shipped in one day, so we divide it by 15 to get LU per hour, rounded up we have 116 LU/HOUR.

• Find the volume of the LU: Knowing the dimensions of FEU (12.2m length with 2.4m width with 2.6m height) we take each value and we divide it by the size of each load unit (1 × 2 × 1 and 0.8 × 1× 0.8).

• How many trucks are needed (product volume/truck volume: We find the Lu/truck and then we divide the number of LU by the Lu/Truck to get the number of trucks per day (rounded up) which rounded up will be equal to **82**.

• Number of trucks per hour: In total we have **82** trucks per day, so we divide it by 15 to get a truck per hour (equals to **6** (rounded up)).

• Number of docks : which is equal to $\frac{Truck}{hour} \times \frac{loadingtime}{unloadingtime}$ rounded up, we get the value 6 which is trucks per hour and we multiply it by ts (loading/unloading time) which is 0.75 hours (45 minutes). The dock number is **5.**

• Receiving area which is equal to: Base area of LU × $\frac{LU}{hour}$ × handling time inside the warehouse (rounded up). Here LU/hour is 61.2, ts = 0.75 hour (same for tm (handling time)). Following the same procedure as in previous steps, we obtain **2** trucks per hour and **2** docks.

• We calculate the area then we divide it by the number of docks and we find the buffer area dedicated to each dock.

For Receiving, we end up with an area of **93 mˆ2** and **3 docks**. We should notice that not all 3 of the docks will be used all days of the week, which means that **31 mˆ2 per dock**. The same procedure is followed for the Shipping area, we end up with an area of **261 mˆ2** and **5 docks** which means buffer area of **52.2 mˆ2** per each dock.

4.6 Number of FLT

The number of forklifts is calculated based on the distance of the traveling of the forklift. The horizontal and vertical distance needed to be taken into consideration. Distance is calculated based on the Centre of Gravity method.

This method considers the shortest and the longest distance that a forklift will travel. Taking a forklift truck from standard, we picked one with a horizontal velocity of 2.91 m/s and a vertical (lifting/lowering) velocity of 0.38 m/s. We need to add a fixed time that a forklift requires due to delays and we are adding 45 seconds for that.

So, the first step is the calculation of the distances. Let's consider the movement of material from the raw materials warehouse to a workstation line.

NOTE: -We are choosing the WORST case scenario for the distances.

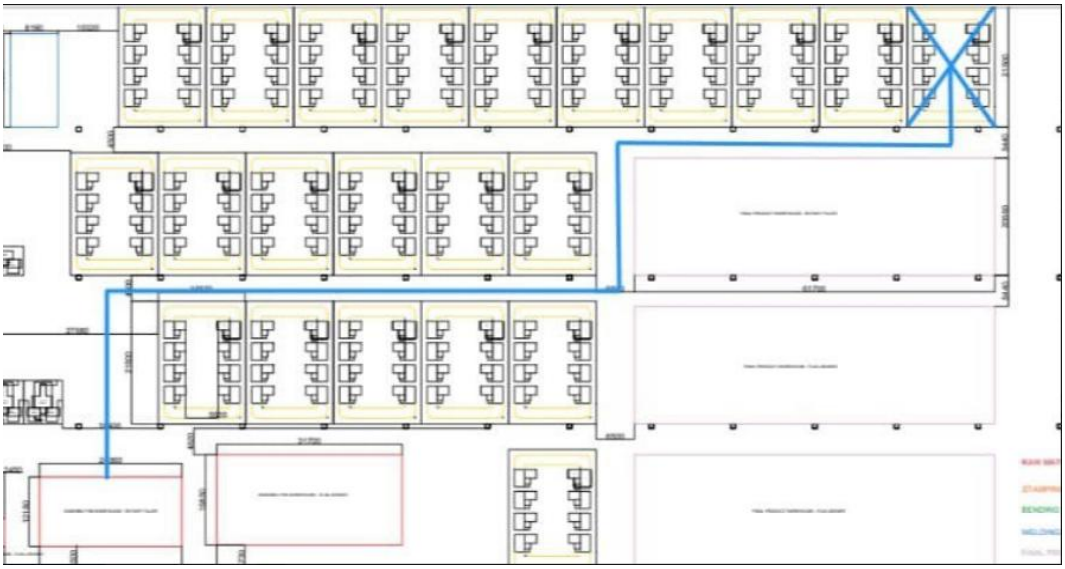
-BLUE line for rotary tiller

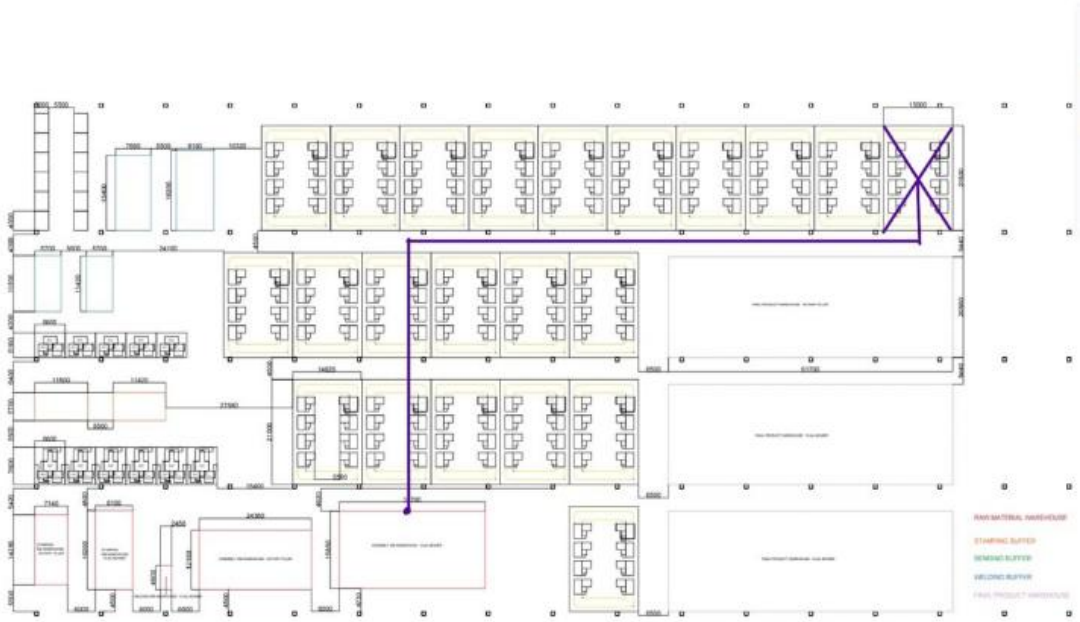
-PURPLE line for flail mower

<u>From raw material warehouse to Final Assembly:</u>

Flail Mower: We must move a number of 2049 LU per month, measuring the distances in vertical and horizontal directions, we get a number of 0.552 forklifts which will be rounded up later.

Rotary Tiller: We must move a number of 1380 LU per month, measuring the distances in vertical and horizontal directions, we get a number of 0.378 forklifts which will be rounded up later.

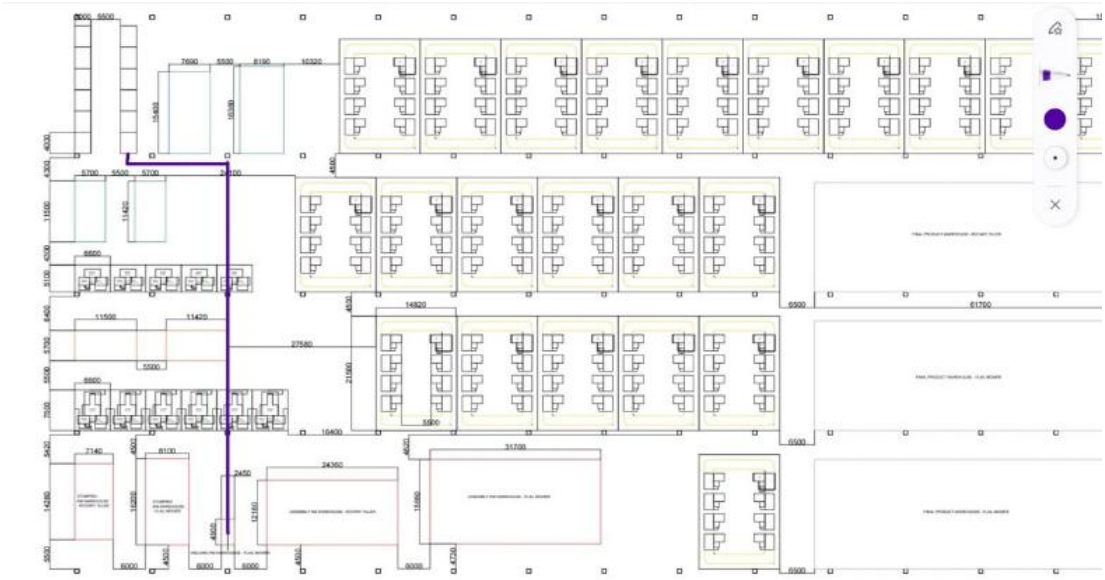




From raw material warehouse to the Welding station:

Flail Mower: We must move a number of 1345 LU per month, measuring the distances in vertical and horizontal directions, we get a number of 0.255 forklifts which will be rounded up later.

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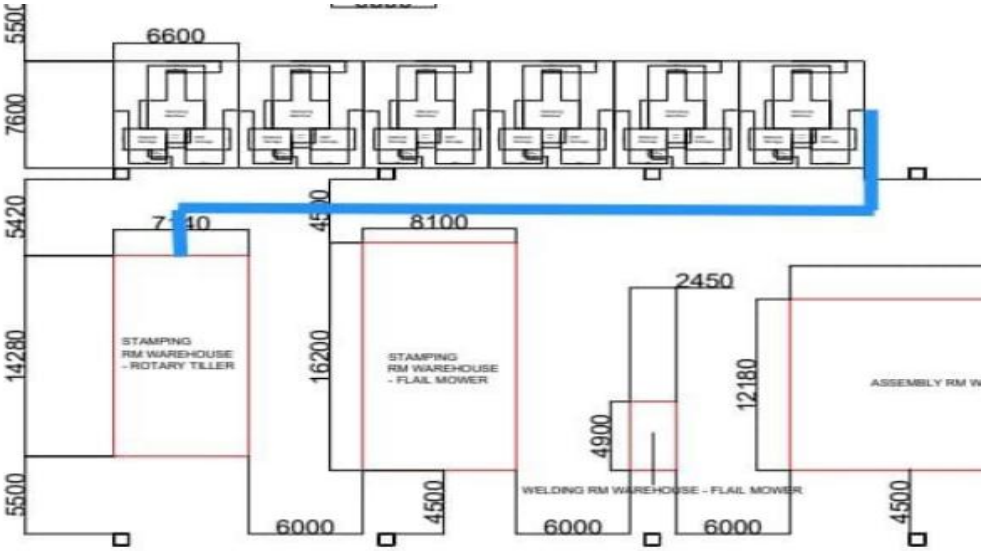


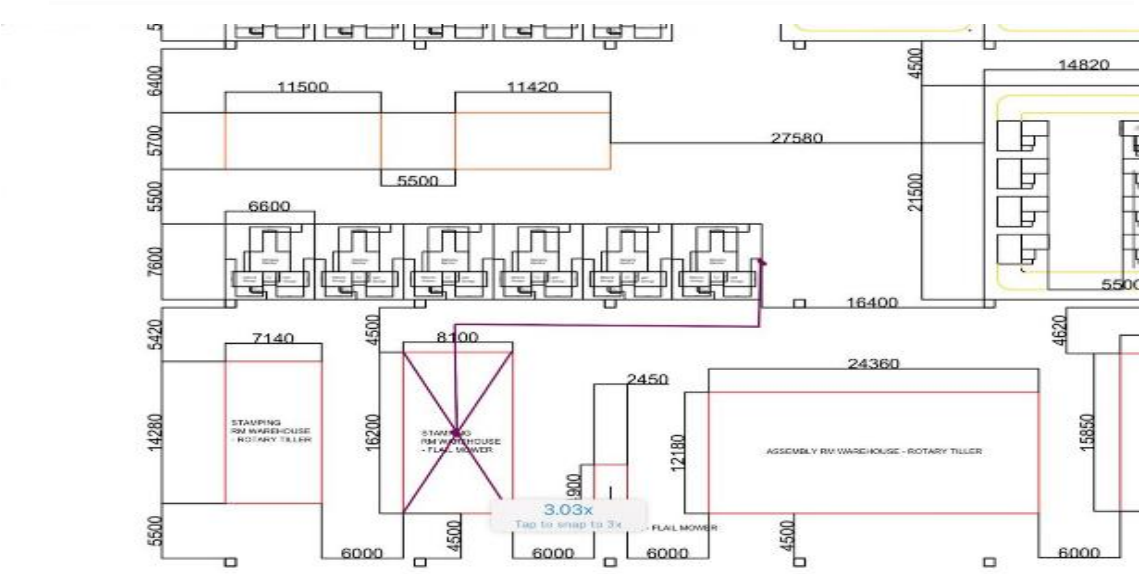
From raw material warehouse to the Stamping station:

Flail Mower: We must move a number of 507 LU per month, measuring the distances in vertical and horizontal directions, we get a number of 0.078 forklifts which

will be rounded up later.

Rotary Tiller: We must move a number of 393 LU per month, measuring the distances in vertical and horizontal directions, we get a number of 0.061 forklifts which will be rounded up later.



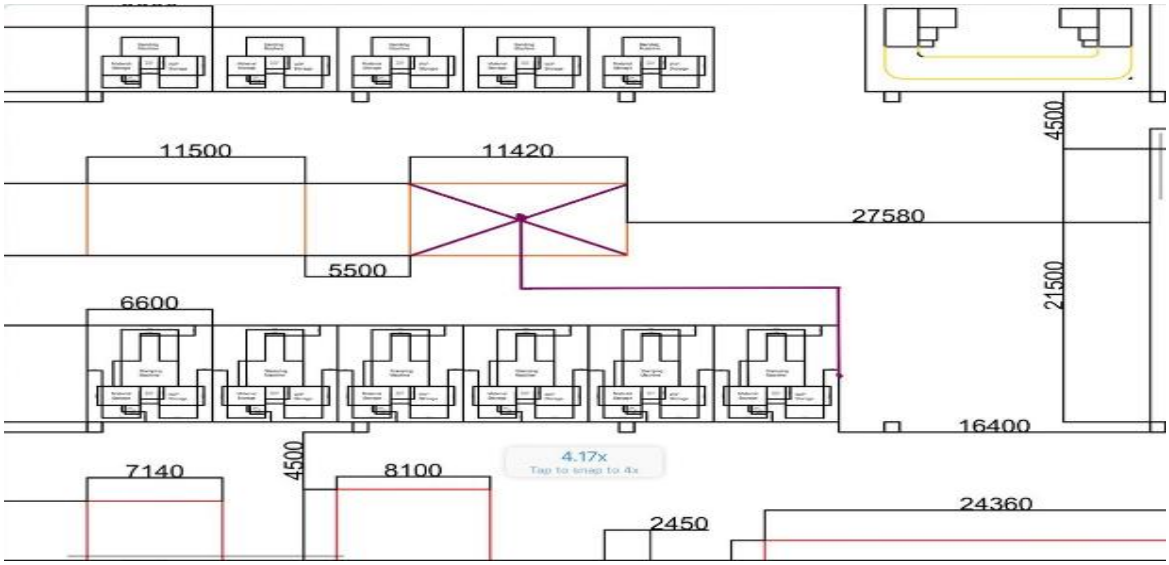


From Stamping to Stamping buffer:

Flail Mower: We must move a number of 1142 LU per month, measuring the distances in vertical and horizontal directions, we get a number of 0.169 forklifts which will be rounded up later.

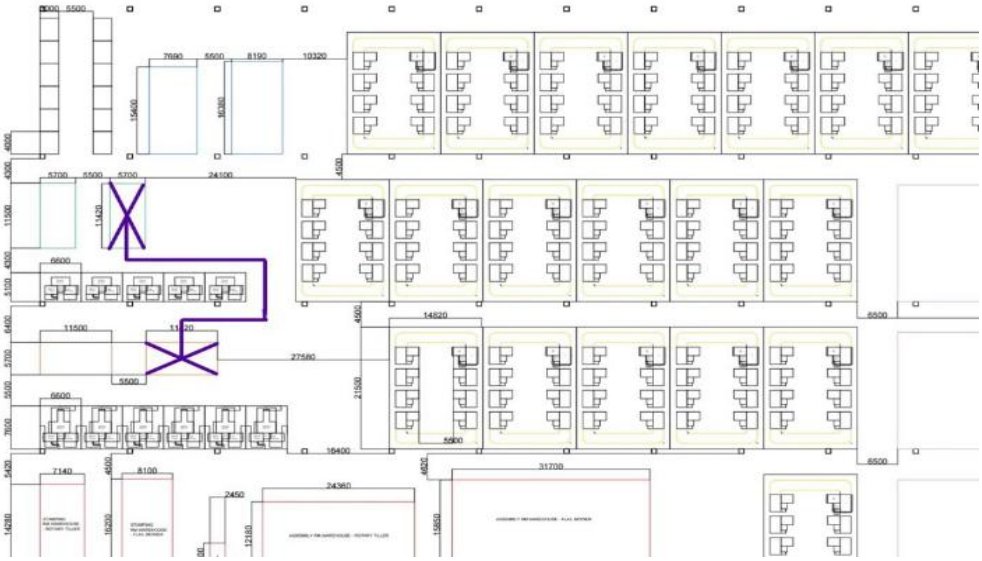
Rotary Tiller: We must move a number of 869 LU per month, measuring the distances in vertical and horizontal directions, we get a number of 0.135 forklifts which

will be rounded up later.



From Stamping buffer to Bending buffer:

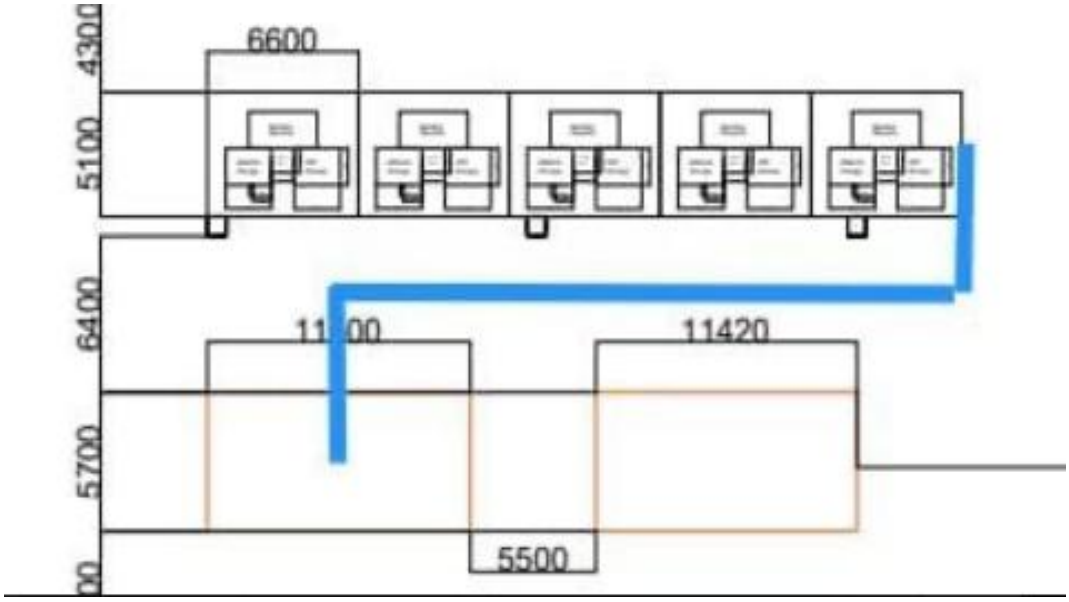
Flail Mower: We must move a number of 1142 LU per month, measuring the distances in vertical and horizontal directions, we get a number of 0.190 forklifts which will be rounded up later.



From Stamping buffer to Bending:

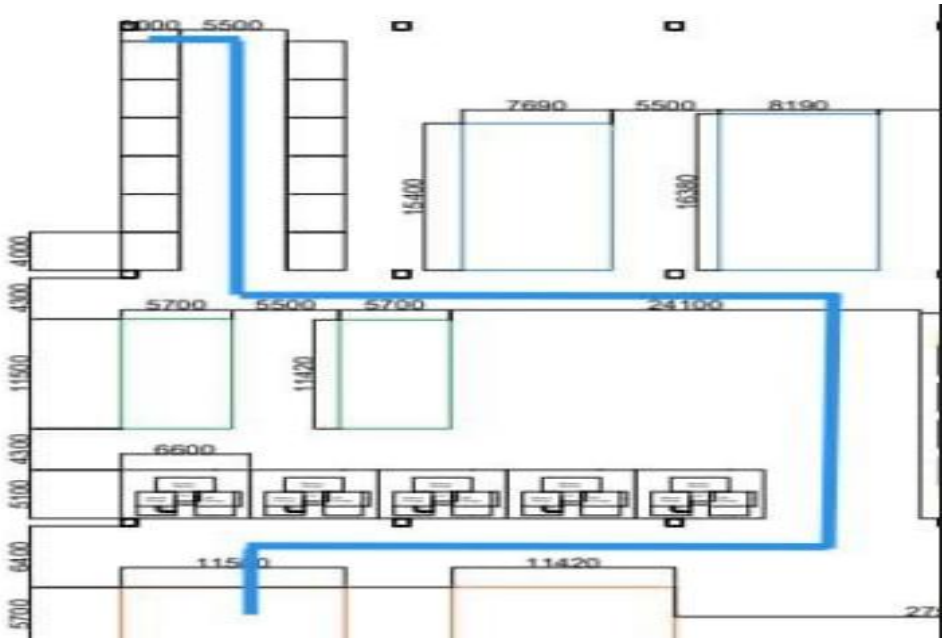
Rotary Tiller: We must move a number of 760 LU per month, measuring the

distances in vertical and horizontal directions, we get a number of 0.112 forklifts which will be rounded up later.



From Stamping buffer to Welding:

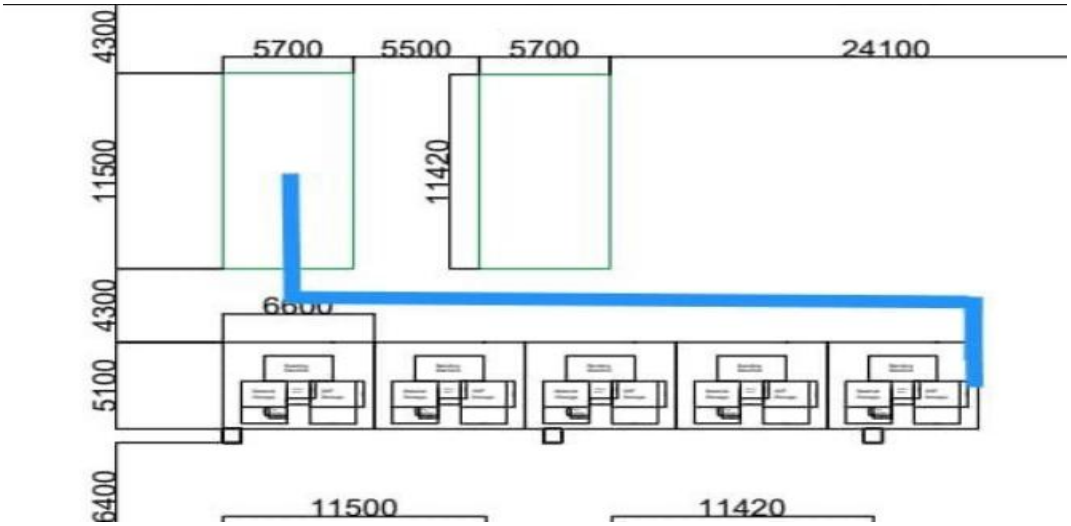
Rotary Tiller: We must move a number of 109 LU per month, measuring the distances in vertical and horizontal directions, we get a number of 0.021 forklifts which will be rounded up later.



From Bending to Bending Buffer:

Rotary Tiller: We must move a number of 738 LU per month, measuring the

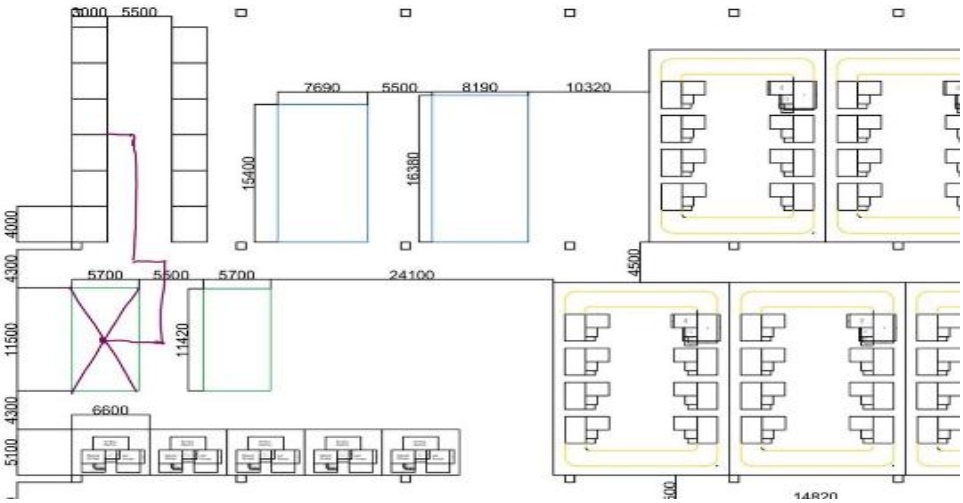
distances in vertical and horizontal directions, we get a number of 0.111 forklifts which will be rounded up later.

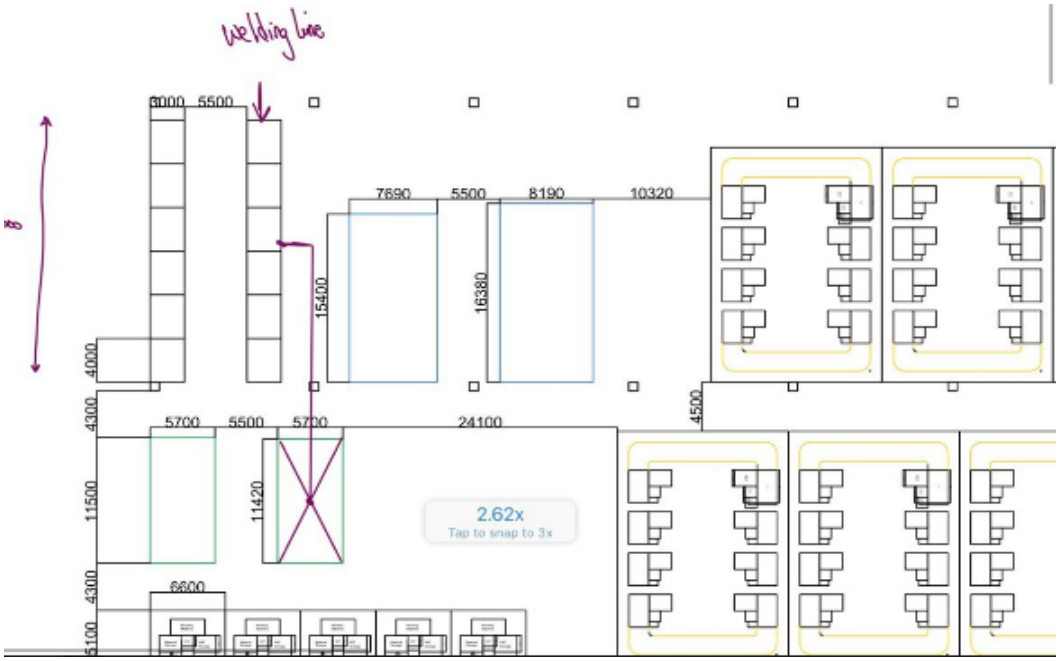


From Bending buffer to Welding:

Flail Mower: We must move a number of 1116 LU per month, measuring the distances in vertical and horizontal directions, we get a number of 0.161 forklifts which will be rounded up later.

Rotary Tiller: We must move a number of 738 LU per month, measuring the distances in vertical and horizontal directions, we get a number of 0.111 forklifts which will be rounded up later.

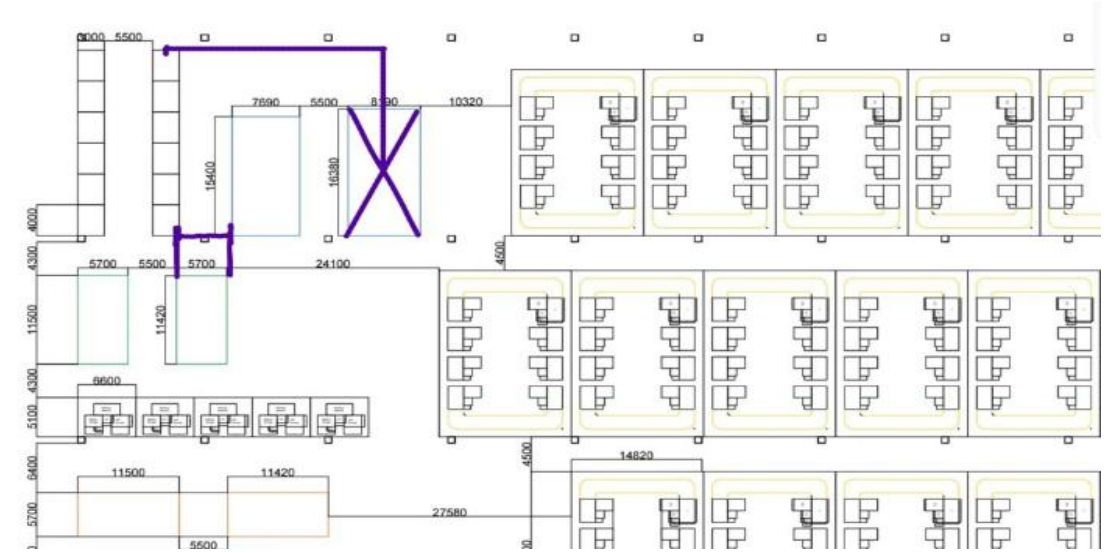


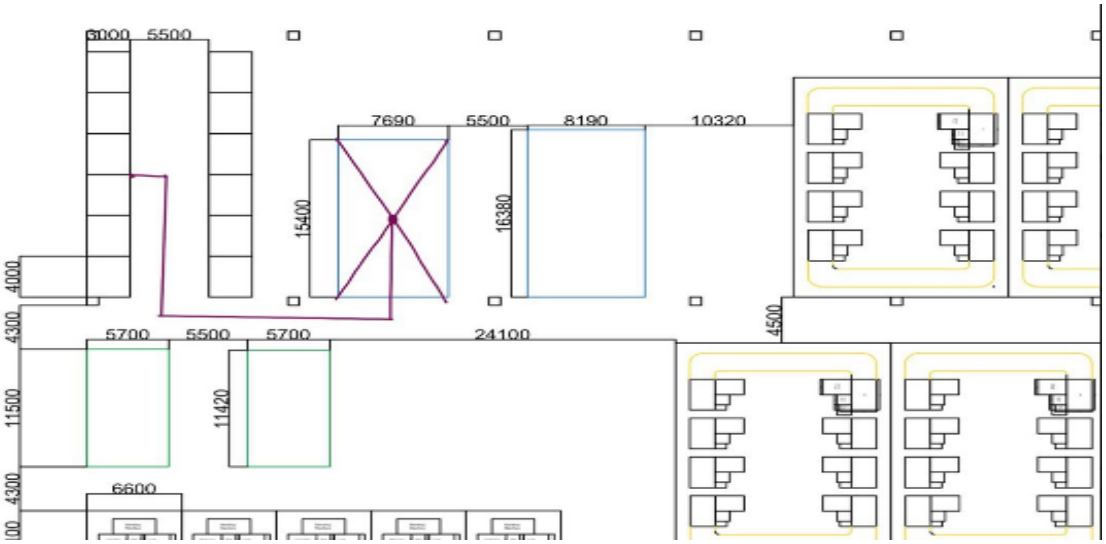


From Welding to Welding buffer:

Flail Mower: We must move a number of 2461 LU per month, measuring the distances in vertical and horizontal directions, we get a number of 0.388 forklifts which will be rounded up later.

Rotary Tiller: We must move a number of 847 LU per month, measuring the distances in vertical and horizontal directions, we get a number of 0.128 forklifts which will be rounded up later.



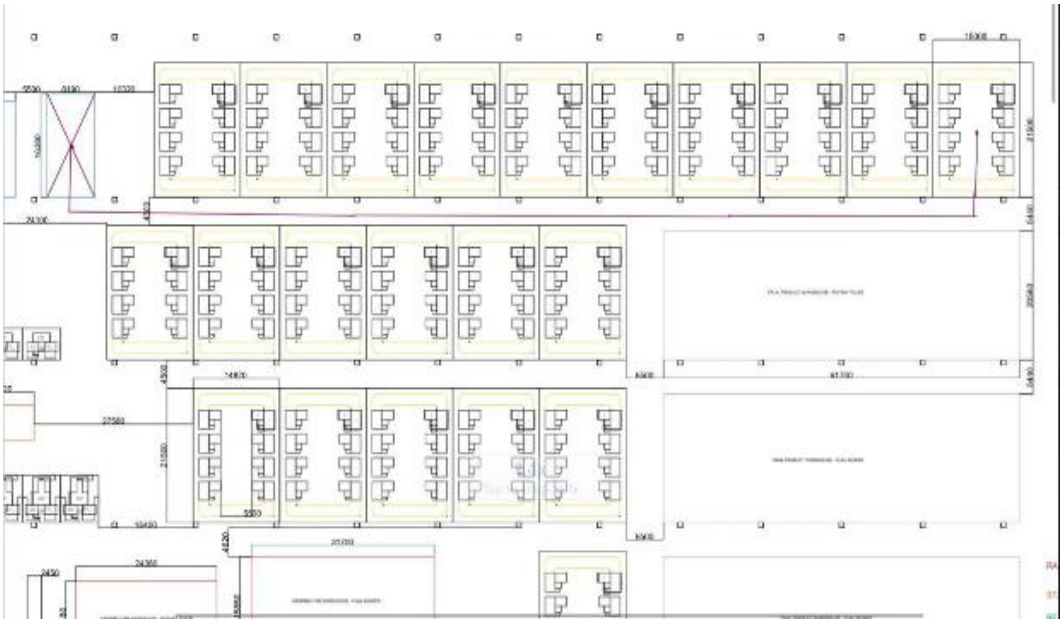


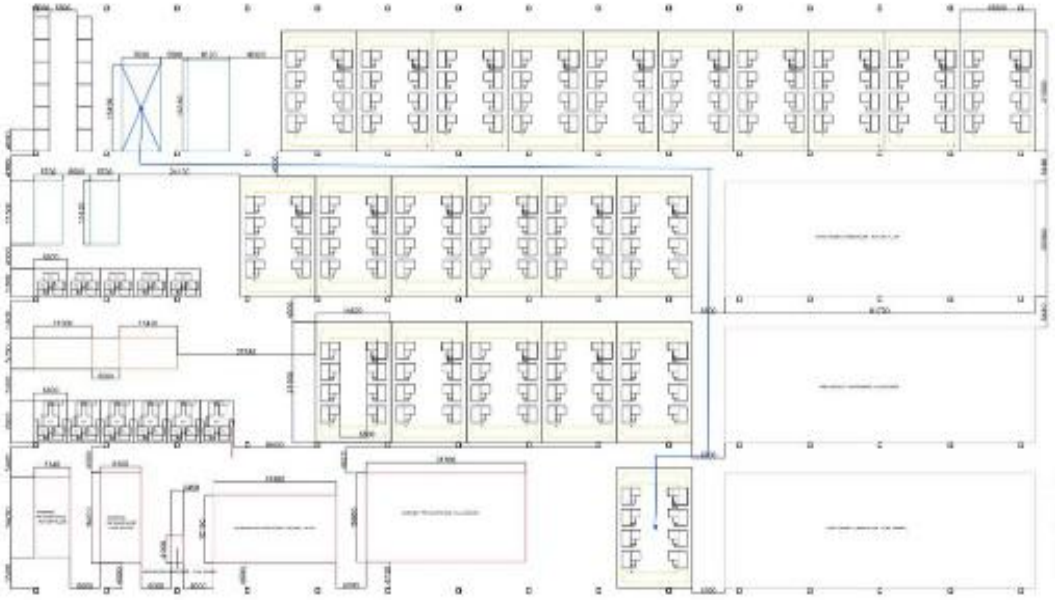
From Welding buffer to Final Assembly:

Flail Mower: We must move a number of 2461 LU per month, measuring the distances in vertical and horizontal directions, we get a number of 0.636 forklifts which will be rounded up later.

Rotary Tiller: We must move a number of 847 LU per month, measuring the distances in vertical and horizontal directions, we get a number of 0.234 forklifts which

will be rounded up later.

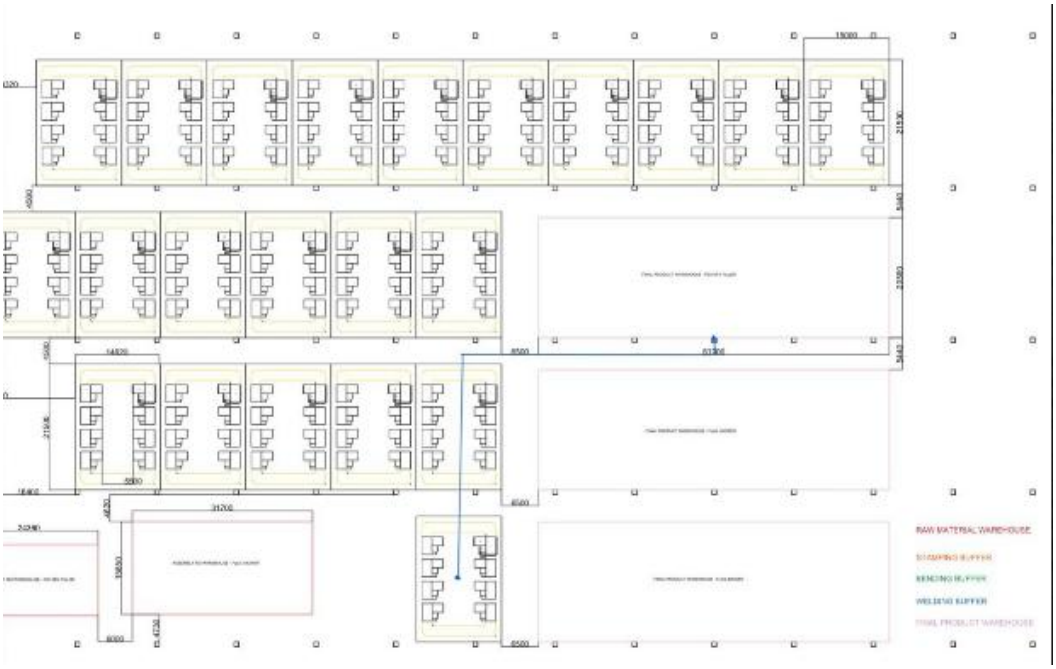




From Final Assembly to FULL PRODUCT warehouse:

Flail Mower: We must move a number of 4510 LU per month, measuring the distances in vertical and horizontal directions, we get a number of 1.039 forklifts which will be rounded up later.

Rotary Tiller: We must move a number of 2227 LU per month, measuring the distances in vertical and horizontal directions, we get a number of 0.422 forklifts which will be rounded up later.



In the end, for the Flail Mower, we ended up at a number of **4 FLT** required and **2 FLT** for

Rotary tiller. Regarding the shipping and receiving docks, we calculated **2 FLT** for the flail mower and **1 FLT** for the rotary tiller , so we resulted in a total of **9 FORKLIFTS** overall .

4.7 Direct and Indirect Staff

For the whole stamping line, which includes both Rotary Tiller and Flail Mower, we have 6 workers per shift so in total 12 workers for the 2 shifts, for bending we have 5 workers per shift, for welding we have 13 workers per shift, in the assembling stations we have the highest number of workers which is 22 workers per shift. For transportation of the products, since we have 9 Forklifts so we need 9 forklift drivers per shift and also we need 8 hand truck operators per 1 shift. In addition, we need 8 supervisors for each shift to help lead and guide the team of workers, ensuring the tasks are being completed efficiently. Lastly we have 15 Employees per shift. Overall, in our plant we relied on a total of **172 staff members** of variant roles divided into 2 shifts.

This table summarizes the distribution above:

Number of Workers	
Forklift Drivers	9
Hand Truck Operators	8
Total Stamping Stations	6
Total Bending Stations	5
Total Welding Stations	13
Total Assembling Stations	22
Total Supervisors 1 shift	8
Total Supervisors 2 shifts	16
Total Workers 1 shift	71
Total Workers 2 shifts	142
Total Employees 1 shift	15
Total Employees 2 shifts	30
Total Staff	172
Total Staff - 1shift	86

Table 7:Number of workers

5 - Safety Design

The safety analysis will be carried using NIOSH to determine whether the working conditions in the welding and assembly departments will ensure worker protection. Specifically, the focus will be on highlighting these cases, which will be the main concern of our study.

-The worker in charge assembles the base structure of the Flail Mower by placing the frame, fastening the gearbox and register hammer cover, and finally laying the complete unit into the load unit for installation.
-The worker in charge safeguards the motor and prepares the Flail Mower for further assembly by mounting the hammer shaft, assembling the drive shaft, and attaching the side gear cover to the frame.
-The worker in charge enhances the functionality and usability of the Flail Mower by fastening scrapers to the gearbox, assembling the frame side closure, installing the bevel gear pair, and securing the drive shaft supports to the chassis.

-The worker in charge ensures proper functioning of the cutting components by assembling the hammers cover onto the frame and mounting the gears onto the frame side closure.
We have different multipliers taking into account to calculate the Recommended Weight Limit(RWL) the initial and final height, traveled distance, frequency of lifting, quality of the coupling and horizontal distance from the picked part. To sum thing sup, the following equation is*Type equation here.* used:

$$\text{RW L} = \text{HM} * \text{V M} * \text{DM} * \text{AM} * \text{CM} * \text{FM} * \text{LC}$$

where LC is taken as a standard 25Kg for a healthy man between the age of 18 and 44 RWL is the recommended weight limit
And thus the lifting index criteria can be calculated to assess the risk as follows

$$LI = \frac{L}{RWL}$$

Where L is load to be lifted by the worker
Moving on to handling welded frames into LU. After a preliminary evaluation of different multipliers we have found out that the Lifting Index is higher that 1 indicating an unsafe task. However such risk is overcome by adding another operator to the Picking Point for the flail mower welded frame assembly, reducing the frequency of lifting per worker and achieving the desired LI less than 1 to guarantee we are in safe zone.
In calculating VLI parameters, a frequency independent recommended weight limit and a frequency independent lifting index are driven for each task according to the following

and $FIRWL_i = LC * HM_i * VM_i * DM_i * AM_i * CM_i$

$$FILI_i = \frac{L_{maxi}}{RIRWL_i}$$

Subsequently, the RWL and LI of each task will be calculated as before. Such data will be used to fill out the a table to calculate the final Variable Lifting Index according to:

$$VLI = LI_1 + \sum \Delta FILI_{2-n}$$

CALC FIRWL									
owering	LC	VM	VD	HM	AM	CM			
low 1	25	0.82	0.87	0.71	1	0.9			
low 2	25	0.86	0.88	0.71	1	0.9			
low 3	25	0.87	0.88	0.71	1	0.9			
low 4	25	0.91	0.89	0.71	1	0.9			
low 5	25	0.98	0.93	0.71	1	0.9			

Calculating RWL							
owering	FIRWL	FM	RWL	Lmax	FILI	Lmed	LI
low 1	11.50347565	0.81	9.317815276	10.296	0.895033841	10.296	1.104980051
low 2	12.15546728	0.81	9.845928497	10.296	0.847026261	10.296	1.045711433
low 3	12.31250288	0.81	9.973127333	10.296	0.836223155	10.296	1.032374265
low 4	13.03429859	0.81	10.55778186	10.296	0.789915923	10.296	0.975204843
low 5	14.63029195	0.81	11.85053648	10.296	0.703745355	10.296	0.868821426

Calculating VLI								
	Lowering	Lmed	RWL	LI	Fmcumul	FILI	deltya(1/Fm(1,...,n))	deltaVLI
1	Row 1	10.296	9.317815276	1.104980051	0.87	//	//	0.22099601
2	Row 2	10.296	9.845928497	1.045711433	0.81	0.847026261	1.234567901	0.209142287
3	Row 3	10.296	9.973127333	1.032374265	0.68	0.836223155	0.236020334	0.206474853
4	Row 4	10.296	10.55778186	0.975204843	0.48	0.789915923	0.612745098	0.195040969
5	Row 5	10.296	11.85053648	0.868821426	0.37	0.703745355	0.619369369	0.173764285
								1.005418404

6 – Choice of LOCATION

Last step is to find a suitable location for the building of the production plant. We decided to get the components of the Flail Mower and Rotary Tiller from **Kuhn Italia SRL** which takes place in “Via Cerca per Colturano, 8, 20077 Melegnano MI”. Since our supplier is located in Melegnano ,we decided to choose a location near the company .

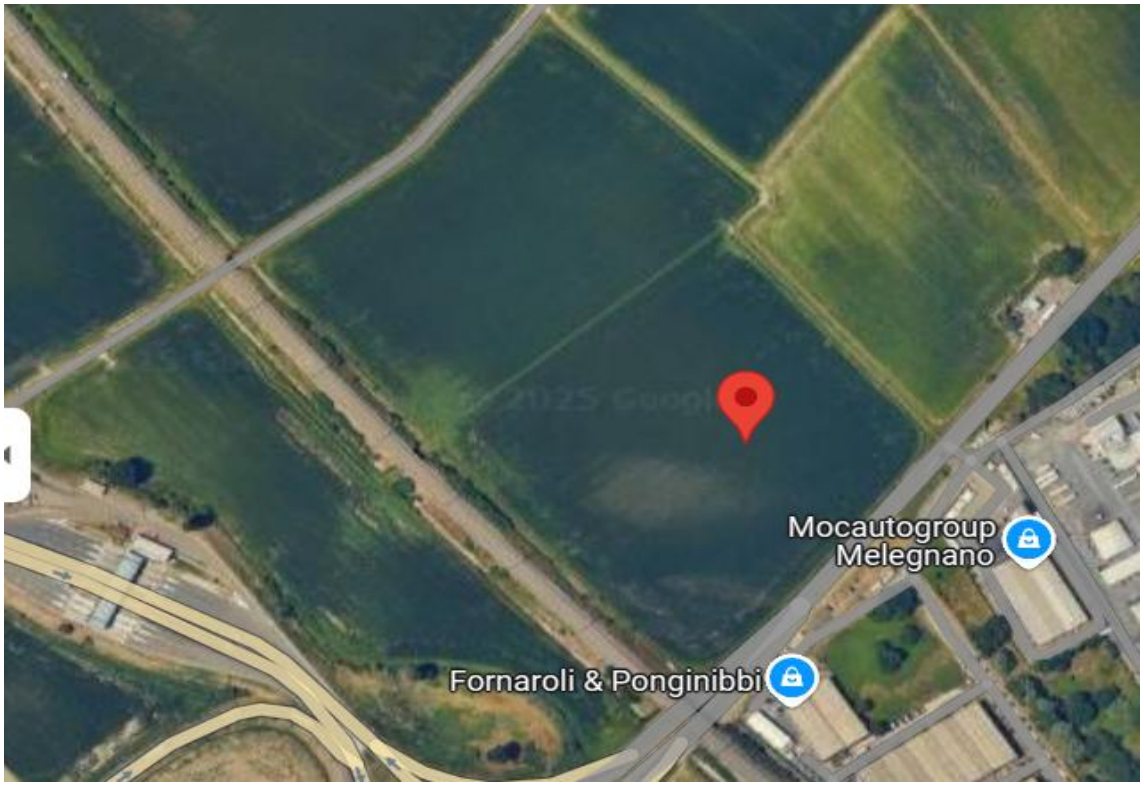


Figure 15 : Location of our company

Another benefit of this location is its **proximity to the supplier**, ensuring that the trucks arrive on time even during traffic delays. Additionally, our L-shaped company layout paired with the two intersecting roads shown in the diagram allows us to allocate one road exclusively for shipping trucks, while the other can be dedicated to incoming deliveries and employee parking.

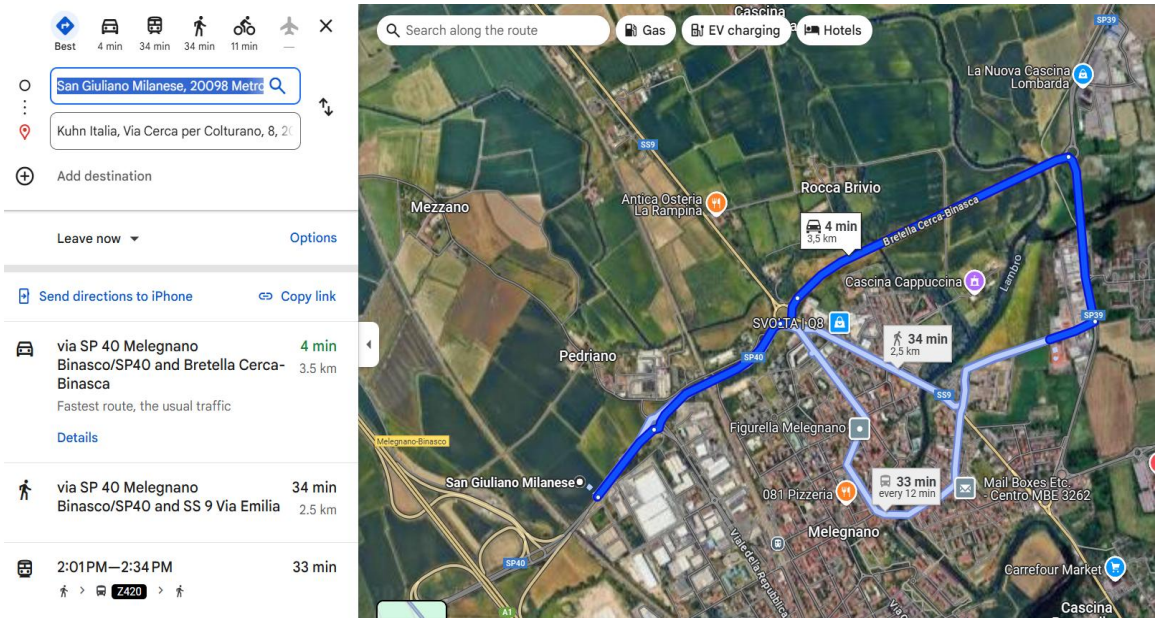
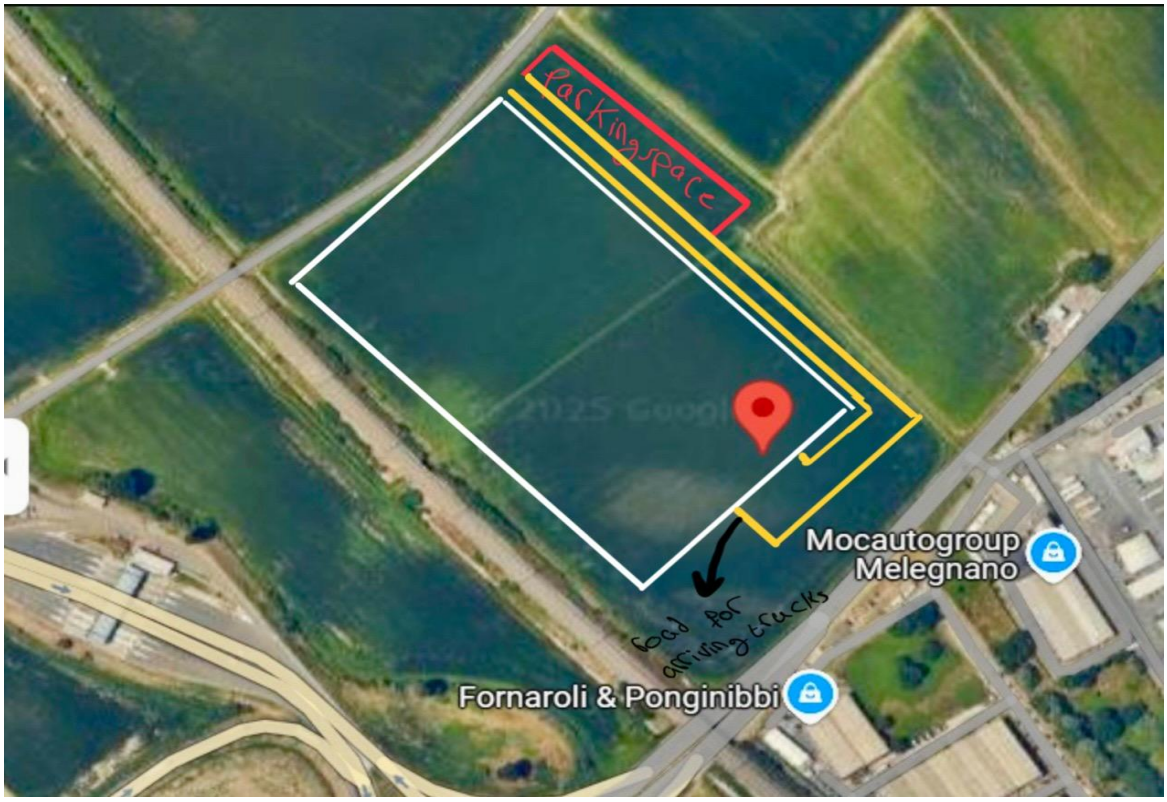


Figure 16:Distance traveled from Kuhn to our plant



7 - Additional considerations in the Layout Plan

In addition to the area of production and storage, the following necessities were taken into account:

• Supervision offices

• Various facilities for the workers, such as changing rooms, canteens, food courts, etc

• Bathrooms

• Necessary maintenance rooms

• Truck parking lots/waiting areas in case all the docks are full

• Employees' parking lots

• Forklift truck charging station

- Main entrance

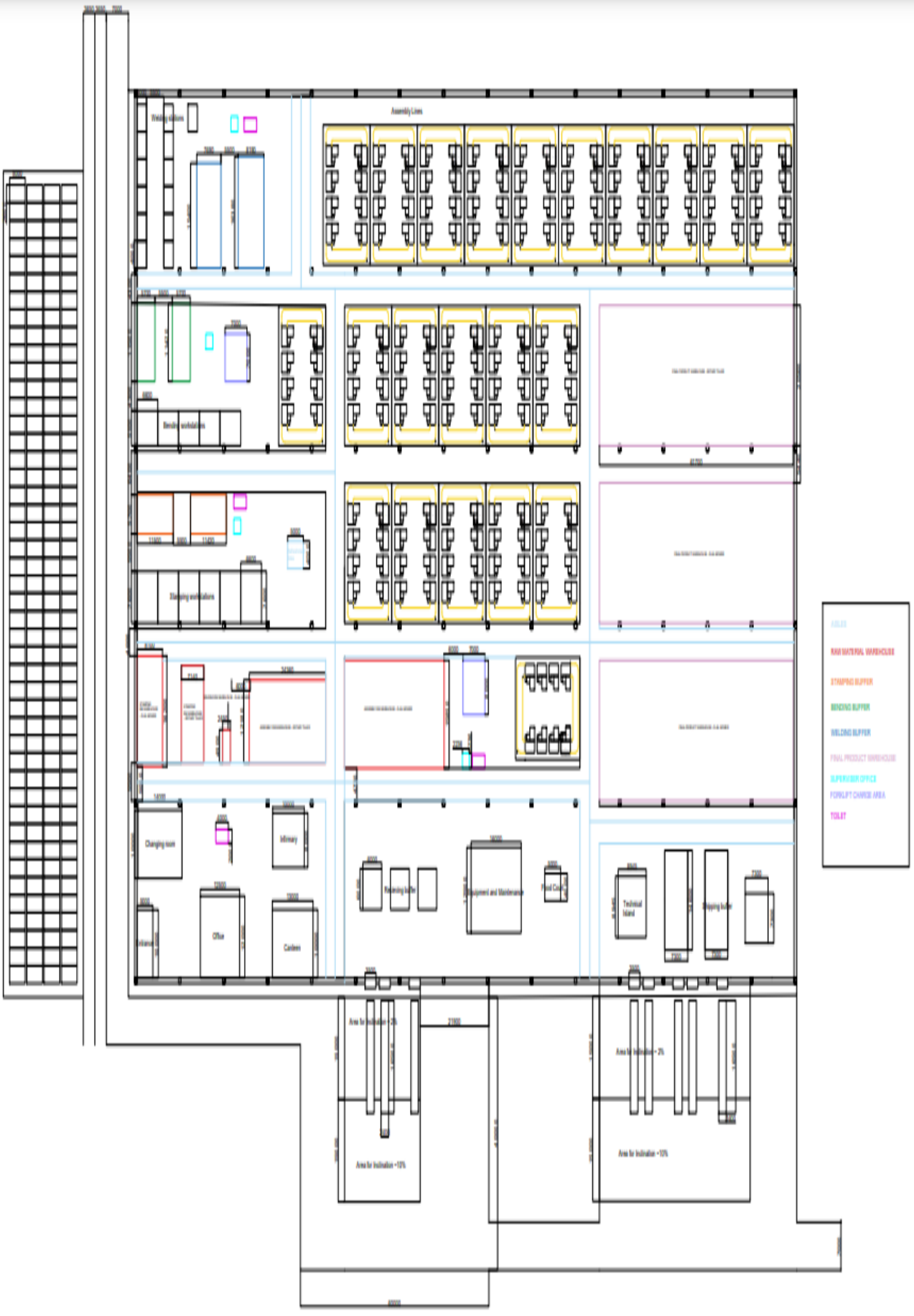

Assembly Lines
Welding workstations
Bending workstations
Stamping workstations
Changing room
Office
Canteen
Primary
Reclaiming buffer
Equipment and Maintenance
Final Control
Technical Island
Shipping buffer
Area for inclination = 3%
Area for inclination = 10%
RAW MATERIAL WAREHOUSE
STAMPING BUFFER
BENDING BUFFER
WELDING BUFFER
FINAL PRODUCT WAREHOUSE
SUPERVISOR OFFICE
FORKLIFT CHARGE AREA
TOILET

Figure 17 ::Layout design with parking space for workers and trucks(receiving and shipping)